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SUPERSEDING
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17 November 1981

MILITARY SPECIFICATION

MULTIPLEXER, TD-1235(P)/TTC

This specification is approved for use by the Communications-Electronics Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, fabrication and testing necessary for the manufacture of the Multiplexer, TD-1235(P)/TTC. This unit is one of the electronic units that comprise the Digital Group Multiplexer (DGM) family of cable system equipment developed by the Joint Tactical Command, Control and Communications Agency (JTC^3A formerly TRI-TAC). See 6.1 for intended use and functional description of the unit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Communications-Electronics Command, ATTN: AMSEL-ED-LC, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-E-55585	Electronics Equipment and Parts, Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461A Notice 4	Electromagnetic Interference Characteristics Requirements for Equipment, Subsystem and System
MIL-STD-462 Notice 3	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781C	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-810D	Environmental Test Methods and Engineering Guidelines
MIL-STD-1332	Definitions of Tactical, Prime, Precise, and Utility Terminologies for Classification of the DOD Mobile Electric Power Engine Generator Set Family
MIL-STD-2073-2	Packaging Requirements Code
MIL-STD-45662	Calibration System Requirements

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

SPECIFICATIONS

Communications-Electronics Command

CR-CX-0173-001	Nuclear Survivability/Vulnerability Provisions for the Family of Digital Group Multiplexer Equipments
CR-CX-0174-001	TEMPEST Provisions for the Family of Digital Group Multiplexer Equipments

DRAWINGS

Communications-Electronics Command

DL-SM-B-876630	Multiplexer, TD-1235(P)/TTC
SM-D-884567	Circuit Card Assembly (CCA), Multi-Vdc
SM-D-884571	CCA, 52 Vdc
SM-D-884579	CCA, AC Input
SM-D-959323	CCA, Analog Applique Unit (AAU)
SM-E-986235	CCA, Loop Modem
SM-E-986239	CCA, Timing Generator
SM-E-986241	CCA, BITE 2
SM-E-986244	CCA, Interface
SM-E-986247	CCA, MUX/DEMUX
SM-E-986250	CCA, Group Framing

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article units. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction.

3.2.1 Fabrication. The TD-1235 shall be constructed in accordance with the requirements of this specification and of DL-SM-B-876630 including the requirements for parts, materials and processes therein.

3.2.2 Printed wiring boards. The printed wiring boards shall meet the requirements of MIL-STD-454, Requirement 17.

3.2.3 Semiconductor devices and integrated circuits. The semiconductor devices and integrated circuits shall meet the requirements of MIL-STD-454, Requirements 30 and 64.

3.3 Parts, materials, and processes-general. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein shall apply.

3.4 Finish. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawing DL-SM-B-876630.

3.5 Marking. The marking shall conform to MIL-M-13231, with approvals in accordance with requirements specified in the contract or purchase order.

3.5.1 Serial numbers. (See 4.15) A unique serial number is required on each nomenclatured item and CCA. The numbers shall be consecutively assigned.

3.6 Electrical preconditioning (burn-in). (See 4.6). Prior to submission to the specified quality conformance inspections, each TD-1235 assembly shall be electrically preconditioned for a minimum of 48 hours operating time. The last 24 hours of electrical preconditioning shall be failure free. Each unit fabricated shall be subjected to this process. Failure free is defined as (1) no built-in test equipment (BITE) alarms during burn-in and (2) the successful completion of the operational test of 4.7.11 after burn-in.

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3.7 Electrical. (See 4.7)

3.7.1 Balanced equipment-side signal characteristics. (See 4.7.1)

3.7.1.1 Line-to-line voltage. (See 4.7.1.1) The line-to-line voltage shall be between +0.1 and +4.0 volts for a digital 1 and between -0.1 and -4.0 volts for a digital 0, measured into a 100 ohm dummy load.

3.7.1.2 Line-to-ground voltage. (See 4.7.1.2) The magnitude of the line-to-ground voltage shall not exceed 3 volts.

3.7.1.3 Clock/data phasing. (See 4.7.1.3) For those data signals having a related clock signal, the time difference between the upward transition of the clock signal and either the upward or downward transition of the data signal shall be within ±240 ns for group rate signals. Times shall be measured at the zero voltage point of the signals.

3.7.1.4 Offset. (See 4.7.1.4) The magnitude of the offset voltage measured between the center tap of a 100 ohm dummy load and ground shall not be greater than 3 volts.

3.7.1.5 Signal balance. (See 4.7.1.5). The magnitude of the difference between the magnitudes of the digital 1 level and the digital 0 level shall be less than 10% of the larger of the two.

3.7.1.6 Slew rate. (See 4.7.1.6) When 100 feet of RG-108A/U cable terminated in a 78 ohm dummy load is connected to the driver output, the transition between ±40 mV (either direction) shall take no more than 10 ns for group rate signals.

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3.7.1.7 Receiver dynamic range. (See 4.7.1.7) The receiver shall deliver the proper digital output when a line-to-line voltage of +50 mV to 4V for a digital 1 or -50 mV to -4V for a digital 0 is applied to the receiver input terminals.

3.7.1.8 Input load impedance. (See 4.7.1.8) The input impedance shall be 100 ohms ±10%.

3.7.1.9 Clock asymmetry. (See 4.7.1.9) The positive half of the clock waveform shall be equal to half the period ±120 ns for group rate timing signals.

3.7.1.10 Timing modes. (See 4.7.1.10)

3.7.1.10.1 Loop back timing mode. Transmit timing shall be derived from the received group signal. The multiplexer shall accept receive group timing frequency errors up to ±50 ppm from the nominal.

3.7.1.10.2 Station clock mode. Transmit timing shall be derived from an external timing source. The received group signal shall be slaved to this timing source by external means. When the external timing signal is absent, the transmit timing shall be derived from an internal timing source. The frequency of this timing source shall be within ± 40 ppm of the nominal frequency.

3.7.1.10.3 External clock/data phasing. The phase relationship between the station clock input or the group related timing input (depending on the timing mode selected) and the group data output shall be such that the negative to positive transition of the input timing shall lead the output data transitions by 0 ± 1/4 of the bit interval.

3.7.2 Digital channel characteristics. (See 4.7.2)

3.7.2.1 Output voltage. (See 4.7.2.1) The output voltage shall be 3 volts ± 15% peak-to-peak measured into a 125 ohm load.

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3.7.2.2 Output waveshape. (See 4.7.2.2) Measured into a 125 ohm resistive load, the rise and fall times (10% to 90%) shall be between 0.65 and 3.0 microseconds. The droop shall be less than 5% of the peak-to-peak amplitude.

3.7.2.3 Input and output impedances. (See 4.7.2.3) The input and output impedance magnitudes shall be 100 to 144 ohms from 8 to 32 kHz and 75 to 144 ohms from 32 to 56 kHz.

3.7.2.4 Input voltages. (See 4.7.2.4) Proper operation shall occur over the input voltage range of 0.15 to 4.0 volts peak-to-peak. Proper operation shall be defined as a bit error rate (BER) of 1×10^{-6} or less.

3.7.3 Loop power feed. (See 4.7.3) Maximum output voltage shall be 56 Vdc into a 11,200 ohm resistive load. Minimum output currents shall be 68.8 mA into a 680 ohm resistive load. The loop outputs shall be short-circuit proof, both line-to-line and line-to-ground. The powerfeed circuitry shall tolerate an initial current surge of 100 mA for up to 45 ms without power feed shutoff.

3.7.4 Idle loop. (See 4.7.4) When transmission from the DSVT has ceased on a loop, the loop modem shall send continuous ones on its equipment-side.

3.7.5 Analog channel characteristics. (See 4.7.5)

3.7.5.1 Analog channel input/output impedances. (See 4.7.5.1) The impedance at the 4 wire analog channel input and output terminals shall be 600 ohms resistive with a minimum return loss of 20 dB over the frequency range 300 to 3400 Hz.

3.7.5.2 Longitudinal balance. (See 4.7.5.2) The longitudinal balance at the analog channel input and output terminals shall be not less than 40 dB when measured at a tone level of 0 dBm over the frequency range of 300 to 3400 Hz.

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3.7.5.3 Frequency response. (See 4.7.5.3) The characteristics of the input/output filters shall be as follows:

Frequency (Hz)	Input Atten (dB)	Output Atten (dB)
76	6 minimum	6 minimum
200	3 ± 1	3 ± 1
1000	reference(0 dB)	reference (0 dB)
3500	3 ± 1	0 to 4
8800	12 minimum	45 minimum

3.7.5.4 Analog channel input and output test tone . (See 4.7.5.4). The nominal test tone input and output level shall be -17 dBm at 820 Hz.

3.7.5.5 Envelope delay distortion. (See 4.7.5.5) The envelope delay relative to 800 Hz of the input and output filters added together shall be less than 100 us from 800 to 3000 Hz.

3.7.5.6 Input-output linearity. (See 4.7.5.6) The output level of each channel shall be within ±1.0 dB of the input signal level from full load tone (see 3.7.5.10) to 35 dB below full load tone when the input and output level selector switches are set to a sum of 0 dB.

3.7.5.7 Loopback switch. (See 4.7.5.7) A loopback switch shall be provided for each channel. It shall permit the digital output of the encoder to be looped to the digital input of the decoder. The loopback switches shall be operable when the modules are inserted into the TD-1235.

3.7.5.8 Noise. (See 4.7.5.8)

3.7.5.8.1 Idle channel noise. (See 4.7.5.8.1) With one or more analog modules inserted in any TD-1235 and the remaining module positions filled with digital modules and the analog channels operating in the loopback mode, the signal measured on the analog inputs and outputs shall not exceed 33 dBrnC when all analog ports are terminated in 600 ohms and input and output level selector switches are set to 0 dB.

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3.7.5.8.2 Crosstalk level. (See 4.7.5.8.2) The interchannel and intra-channel crosstalk plus noise measured at either the near end or the far end of two AAU channels shall not exceed 34 dBrnC when sending the 820 Hz test tone at full load.

3.7.5.9 Peak clipping. (See 4.7.5.9) For the +4 dBm input signal at 200 Hz, the output signal shall have a total distortion of less than 10% at a channel rate of 32 kb/s with the input level selector switch set to -10 dB and the output level selector switch set to +10 dB.

3.7.5.10 Full load tone. (See 4.7.5.10) The nominal full load tone input and output level shall be 0 dBm at a frequency of 820 Hz with the input level selector switch set to -10 dB, and the output level selector switch set to +10 dB. With a full load tone, the Continuous Variable Slope Delta (CVSD) converter approaches the slope overload condition.

3.7.5.11 Analog channel signal processing. (See 4.7.5.11)

3.7.5.11.1 Clocking rate. (See 4.7.5.11.1) The channel clocking rate of the CVSD converter shall be 16 or 32 kb/s, selectable.

3.7.5.11.2 Dynamic range. (See 4.7.5.11.2) The dynamic range of the CVSD converter shall be such that the repetitive test patterns shown below shall produce 666 2/3 Hz tone outputs with a ratio of 23 ± 1 dB.

Channel Rate kb/s	Digital Pattern (Hexademical Equivalent)*											
32 low output	D	D	5	2	4	9	2	2	A	D	B	6
high output	A	E	F	F	F	B	5	1	0	0	0	4
16 low output	6	E	A	9	2	5						
high output	B	F	D	4	0	2						

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3.7.5.11.3 Output level control. (See 4.7.5.11.3) The output level of the CVSD decoder shall be selectable in four steps independent of the level setting at the encoder. The output level selector shall be labeled -3, 0, +10 and +20. With either of the two repetitive digital patterns (shown below) applied to the decoder, the analog output into a 600 ohm load shall be -20, -17, -7, or +3 dBm ± 0.5 dB, when the output level selector is set at -3, 0, +10, or +20, respectively.

Channel Rate kb/s	Digital Pattern (Hexadecimal Equivalent)*
32	F D A A 1 0 2 5 5 E
16	F B 4 1 2

* The serial bit patterns are read from left to right and are expressed as hexadecimal characters; each character is transmitted in time sequence as read, MSB first and LSB last.

3.7.5.11.4 Input level control (See 4.7.5.11.4). The operating level of the CVSD encoder shall be adjustable in four steps. The input level selector shall be labeled +3, 0, -10 and -20. When the channel is in the loopback mode, the input levels and level settings shall produce the output levels shown below with a tolerance of ± 1.0 dB.

Input Level (dBm)	Encoder Level Setting	Decoder Level Setting	Output Level(dBm)
-20	+3	+20	+3
-17	0	+20	+3
-7	-10	+20	+3
+3	-20	+20	+3
-20	+3	+10	-7
-17	0	+10	-7
-7	-10	+10	-7
+3	-20	+10	-7
-20	+3	0	-17
-17	0	0	-17
-7	-10	0	-17
+3	-20	0	-17
-20	+3	-3	-20
-17	0	-3	-20
-7	-10	-3	-20
+3	-20	-3	-20

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3.7.6 Trunk Encryption Device (TED) resync command voltage. (See 4.7.6) The transmit voltage shall have an idling level line-to-line between -0.3 and -3.0 volts. When resync is necessary it shall rise to between +0.3 and +3.0 volts line-to-line. These voltages shall be measured into a 6 kilohm load.

3.7.6.1 TED resync command waveshape. (See 4.7.6.1) The TED resync command pulsewidth shall be greater than 10 us and less than 100 us. The transmit waveform shall be specified into a 10 kilohm load. The signal rise and fall times shall be less than 3 us measured between the 10% and 90% levels.

3.7.7 Frame squelch. (See 4.7.7) This interface shall be an unbalanced TTL signal. It shall be arranged so that if nothing is connected, the frame search is not squelched. The input impedance shall be 1 kilohm minimum, 51 kilohm maximum. The internal pull-up voltage shall be between +4.5 and +5.5 volts. When the voltage at the input is less than +0.8 volt the frame search shall be squelched. When the voltage at the input is more than +2.0 volts the frame search shall not be squelched.

3.7.8 Framing. (See 4.7.8)

3.7.8.1 Format. (See 4.7.8.1) The overhead channel shall be channel 0 of the TD-1235 group. The overhead channel shall be composed of eight equal-rate subchannels as shown in figure 1. When the voice digitization rate is 32 kb/s, the overhead subchannel rate shall be 4 kb/s. When the voice digitization rate is 16 kb/s, the overhead subchannel rate shall be 2 kb/s. Subchannel 0 shall be the frame subchannel. Subchannel 7 shall be the telemetry subchannel. Subchannels 1-6 shall be unused and set to the "ones" state.

3.7.8.2 Frame acquisition time. (See 4.7.8.2) The acquisition time is defined as the period from the start of frame search until the multiplexers at both ends of a link are capable of restoring traffic in both directions on the link between them. The TD-1235 shall acquire frame synchronization within 50/100 ms at frame rates of 4/2 kb/s, with a minimum probability of 0.9, in an error environment of 0.1% BER random errors. The TD-1235 shall also acquire frame sychronization within 100/150 ms at frame rates of 4/2 kb/s, with a minimum probability of 0.9, in an error environment of 20% BER random bursts interspersed with 0.1% BER random error intervals. The time between the start of successive bursts shall be random

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between 50 ms and 1 s, and the burst length shall be 5% of the time between the start of that burst and the start of the next burst. The above times do not include cryptographic resync delays, or any preset intended to assist in flywheeling through fades.

3.7.8.3 Frame maintenance. (See 4.7.8.3) While bit count integrity is maintained, "out-of-sync" shall not be declared more often than once per 240 hours (on the average) in an error environment of 20% BER bursts interspersed with 0.1% BER random error intervals. The time between the start of successive bursts shall be random between 50 ms and 1 s, and burst length shall be 5% of the time between the start of that burst and the start of the next burst.

3.7.8.4 Detection of out of sync. (See 4.7.8.4) When bit count integrity is lost and with random data on all channels, the TD-1235 shall detect out of sync within 150 ms at frame rates of 4/2 kb/s, with a minimum probability of 0.9.

3.7.8.5 Delayed frame search. (See 4.7.8.5) A fixed delay of 500 $\pm$ 50 ms shall be provided.

3.7.8.6 Idle filler pattern. (See 4.7.8.6) When the TD-1235 is out of frame, it shall set its channel outputs to all ones.

3.7.9 Prime power interface. (See 4.7.9) The ac primary power source shall be 115 volts $\pm$10%, single phase. The frequency of the ac voltage shall be 50 Hz -5% to 400 Hz +5%. The ac primary power shall conform to MIL-STD-1332 for Type 1, Class 1 and 2, Mode I, II, and III generator sets. The input power to the TD-1235 shall not exceed 190 watts.

3.7.10 BITE and alarms. (See 4.7.10) The BITE shall monitor signals in and out of the unit and internal circuits. Go-no-go fault-location circuits shall isolate failures to a particular CCA or lowest replaceable unit (LRU). Indicating devices shall be provided to ensure proper operation of the equipment and to monitor the dc voltages, go-no-go fault isolation, and other critical internal and external signals. The BITE shall operate without interrupting traffic or interfering with the normal system operation. Removal of the BITE card shall not affect operation of the unit. Failure alarms shall be provided for channel failure, common power supply failure, and interunit signal malfunctions. Means shall be provided in the unit to

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check the operation of the alarms without interrupting the equipment operation. Failures detected by the BITE circuitry cause relay contact closures accessible externally at a connector on the rear panel. The S and SA alarm outputs as described below are available.

a. S alarm. The S alarm shall indicate any failure detected including loss of input power. This alarm shall also provide a visible indication on the front panel except when there is a loss of input power.

b. SA alarm. The SA alarm shall indicate that the malfunction is within the unit or the unit has lost input power. If the unit power switch is turned off, there shall be no SA alarm.

3.7.11 Operational tests. (See 4.7.11) Each fabricated and tested unit shall be subjected to a final operational test in accordance with 4.7.11 after all other inspections and test are completed to determine that the unit is operating prior to packaging, and meets the requirements 3.7.11.1 through 3.7.11.5.

3.7.11.1 Digital channel check. (See 4.7.11.1) There shall be no errors for a period of 30 seconds when tested in accordance with 4.7.11.1.

3.7.11.2 Alarm monitor check. (See 4.7.11.2) There shall be no alarms generated when the unit is operated normally. The S (but not SA) alarm shall be actuated when the group cable is removed from the unit under test (UUT). The alarm state is defined as a closed set of relay contacts in each of the S and SA connections.

3.7.11.3 Sync/frame check. (See 4.7.11.3) The TD-1235 shall generate a Trunk Encrypton Device (TED) resync command whenever the TD-1235 is out-of-frame and attempting to resynchronize. The TD-1235 shall cease sending the pulse when the unit is in-frame.

3.7.11.4 DC voltage check. (See 4.7.11.4) The TD-1235 digital channel outputs, when enabled, shall provide power to the Digitial Secure Voice Terminal (DSVT) at a voltage between 46.9 Vdc and 56.0 Vdc.

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3.7.11.5 Analog channel check. (See 4.7.11.5) With both level switches set to 0 dB, the gain shall be 0 ± 1 dB at 820 Hz.

3.7.12 Environmental operational measurements.(See 4.7.12).

3.7.12.1 Digital channel check. (See 4.7.12.1) A count of 0 errors shall be obtained over a measurement period of 30 seconds.

3.7.12.2 Alarm monitor check. (See 4.7.12.2) There shall be no alarms generated, when the UUT is operated normally. Depressing the SUMMARY FAULT switch on the UUT shall cause an S and SA alarm. The alarm state is defined as a closed set of relay contacts in each of the S and SA connections.

3.7.12.3 Sync/frame check. (See 4.7.12.3) Turning off the unit outside the test facility shall cause the UUT to generate a resync command pulse. With power on, no resync command pulse shall be generated indicating that the unit is framed up.

3.7.12.4 DC voltage check. (See 4.7.12.4) The loop power feed voltage at each of the digital channels shall be between 46.9 Vdc and 56.0 Vdc.

3.7.12.5 Analog channel check. With all level switches set to 0, the gain at 820 Hz shall vary no more than ± 1 dB from the room temperature value.

3.8 Service conditions. The operational test of 3.7.12 shall be performed before, during, and after each service condition test unless otherwise specified in the appropriate test paragraph.

3.8.1 Temperature. (See 4.8.1)

3.8.1.1 High temperature. (See 4.8.1.1) The equipment shall be operable without degradation in specified performance at ambient air temperatures as high as +63°C (+145°F), and shall withstand exposure (non-operating) to ambient air temperatures as high as +71°C (+160°F).

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3.8.1.2 Low temperature. (See 4.8.1.2) The equipment shall be operable without degradation in specified performance at ambient air temperatures as low as -31.7°C (-25°F), and shall withstand exposure (non-operating) to ambient air temperatures as low as -57°C (-70°F).

3.8.2 Humidity. (See 4.8.2) The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

3.8.3 Low pressure (altitude). (See 4.8.3) The equipment shall be operable without degradation in specified performance at altitudes up to 10,000 feet above sea level, and shall withstand air transportation at altitudes up to 40,000 feet above sea level.

3.8.4 Dust. (See 4.8.4) The equipment shall withstand, in both operating and non-operating conditions, the effects of dust particles that may accumulate within the shelter assembly or vehicle.

3.8.5 Salt fog. (See 4.8.5) The equipment shall be resistant to the corrosive effects of a salt fog atmosphere.

3.8.6 Fungus. (See 4.8.6) The equipment, in the assembled-and-ready-for-delivery condition, shall provide no nutrients in material, coating or contaminant form or support fungal growth when tested in accordance with 4.8.6. Only inherently fungus resistant grades of materials per requirement 4, MIL-STD-454, shall be used.

3.8.7 Vibration and shock. (See 4.8.7). The equipment shall withstand vibration and shock induced during vehicular transport as part of a mobile assemblage, over all types of roads and cross-country terrain, and shocks encountered during servicing.

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3.9 Electromagnetic interference (EMI). (See 4.9) The TD-1235 shall meet the following emission and susceptibility requirements of MIL-STD-461A, Notice 4:

CE02	CS02	RE02	RS02
CE03	CS06	RE02.1	RS03
CE04			
CE05			

NOTES:

a. RE02.1 shall be applicable over the frequency range of 14 kHz to 1000 MHz.

b. RS03 shall be applicable over the frequency range of 10 kHz to 1000 MHz.

c. Conducted emission and susceptibility requirements are applicable to dc, 60 Hz and 400 Hz power lines.

3.9.1 Bonds and grounds. (See 4.9.1) Bonding and grounding impedance shall not exceed 2.5 milliohms.

3.10 Reliability. (See 4.10) The TD-1235 shall demonstrate a specified mean-time-between-failure (MTBF), θ_o of 4000 hours when in the normal mode of operation.

3.11 Interchangeability. (See 4.11) Like units, assemblies, subassemblies and replaceable parts shall meet the Requirement 7 of MIL-STD-454.

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3.12 Unit subassembly electrical performance. (See 4.12) The unit subassemblies listed below shall meet the requirements specified on their respective test requirement specification (TRS) drawings, as referenced in the following subassembly drawings:

CCA	Drawing
Multi-Vdc	SM-D-884567
52 Vdc	SM-D-884571
AC Input	SM-D-884579
Analog Applique	SM-D-959323
Loop Modem	SM-E-986235
Timing Generator	* SM-E-986239
BITE 2	* SM-E-986241
Interface	* SM-E-986244
MUX-DEMUX	* SM-E-986247
Group Framing	* SM-E-986250

*Drawing contains automatic test equipment (ATE) acceptance procedures and programs.

3.13 Operational test. (See 4.7.11) Each fabricated and tested unit shall be subjected to a final operational test in accordance with 4.7.11 after all other inspections and tests are completed to determine that the unit is operating prior to packaging.

3.14 Systems safety engineering. (See 4.13)

3.14.1 Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirements 1 and 8, shall be applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract.

3.14.2 Rounding exposed edges. Exposed edges and corners shall be rounded sufficiently to preclude puncture or laceration hazards.

3.14.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes/spark gaps, lenses).

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3.15 Workmanship. (See 4.13) The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and with applicable portions of MIL-P-11268 as follows:

a. Adhesives

b. Cables, cords and wires, electrical

c. Wire (harnessing)

d. Plastic materials and parts

e. Rubber

f. Wiring and cabling

g. Connectors (electrical)

h. Printed wiring assemblies

i. Processes

3.16 Nuclear survivability. (See 4.16) The equipment shall satisfy the nuclear survivability requirements stated in Section 3 of CR-CX-0173-001 with the exception of paragraphs 3.3.1.1 (BLAST) and 3.3.1.2 (THERMAL).

3.17 TEMPEST. (See 4.17) The TD-1235 shall be designed to comply with the TEMPEST requirements specified in CR-CX-0174-001.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.3) Does not include packaging.

b. Inspections covered by subsidiary documents. (See 4.4)

c. Quality conformance inspections.

(1) Quality conformance inspections of equipment before packaging. (See 4.5).

(2) Packaging inspection. (See 4.14)

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish a minimum of 20 first article units of the complete TD-1235. The first article quantity shall be as specified in the contract or purchase order.

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4.3.2 First article inspection. The first article inspection shall consist of inspections specified in table I and shall be performed in the order specified herein except that all hardwired and plug-in assemblies shall undergo in-process Group A inspection prior to burn-in.

4.3.3 First article inspection data. The first article inspection test plan and test report(s) shall be as required in the contract or purchase order.

NOTES for table I:

a. The table I inspections 1 through 4, in the order shown, shall be performed on all first article inspection units before subjecting these units to any other inspection requirements specified in the table.

b. The numbers in the unit columns in the table covered by inspections 5 and 6 specify the order of inspections for the indicated unit. An asterisk (*) indicates the use of an auxiliary TD-1235 unit.

c. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

d. All first article units shall be subjected to Group A and Group B inspections.

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TABLE I. First article inspection.

Inspection	Rqmt Para	Test Para	Unit 1	2	3	4	5	6	7	8	9	10	11	12-19	20+
1. Inspections by subsidiary documents	N/A	4.4	Insp to be performed on all units.												
2. Burn-In	3.6	4.6	Insp be performed on all units.												
3. Group A inspection	(See Table II)		Insp to be performed on all units.												
4. Group B inspection	(See Table III)		Insp to be performed on all units.												
5. Group C inspection															
Subgroup 1															
High temperature	3.8.1.1	4.8.1.1	1	*											
Low temperature	3.8.1.2	4.8.1.2	2	*											
Subgroup 2															
EMI	3.9	4.9			1										
Vibration/shock	3.8.7	4.8.7				1	*								
TEMPEST	3.17	4.17											1		
Subgroup 3															
Humidity (Note c)	3.8.2	4.8.2						1			*				
Subgroup 4															
Altitude	3.8.3	4.8.3							1		*				
Dust (Note c)	3.8.4	4.8.4							2						
Salt fog (Note c)	3.8.5	4.8.5							3						
Fungus (Note c)	3.8.6	4.8.6								1					
Subgroup 5															
Line- to-ground voltage	3.7.1.2	4.7.1.2(1)										1			
Clock/data phasing	3.7.1.3	4.7.1.3										2			
Offset	3.7.1.4	4.7.1.4(1)										3			
Signal balance	3.7.1.5	4.7.1.5										4			

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TABLE I. First article inspection (contd).

Inspection	Rqmt Para	Test Para	Unit 1	2	3	4	5	6	7	8	9	10	11	12-19	20+
Slew rate	3.7.1.6	4.7.1.6										5			
Input load impendance	3.7.1.8	4.7.1.8										6			
Clock asymmetry	3.7.1.9	4.7.1.9										7			
Timing modes	3.7.1.10	4.7.1.10										8			
Output waveshape	3.7.2.2	4.7.2.2 (D)										9			
Input/output impedance	3.7.2.3	4.7.2.3 (D)										10			
Loop power feed	3.7.3	4.7.3 (D)										11			
Idle loop	3.7.4	4.7.4 (D) (1)										12			
Analog channel input/ output impedance	3.7.5.1	4.7.5.1 (A)										13			
Longitudinal balance	3.7.5.2	4.7.5.2 (A)										14			
Envelope delay distortion	3.7.5.5	4.7.5.5 (A) (1)										15			
Loopback switch	3.7.5.7	4.7.5.7 (A) (1)										16			
Crosstalk level	3.7.5.8.2	4.7.5.8.2 (A) (1)										17			
Peak clipping	3.7.5.9	4.7.5.9 (A) (1)										18			
TED resync command waveshape	3.7.6.1	4.7.6.1										19			
Frame squelch	3.7.7	4.7.7										20			
Format	3.7.8.1	4.7.8.1 (1)										21			
Frame acquisition time	3.7.8.2	4.7.8.2										22			
Frame maintenance	3.7.8.3	4.7.8.3 (1)										23			
Detection of out-of-sync	3.7.8.4	4.7.8.4										24			
Delayed frame search	3.7.8.5	4.7.8.5										25			
Idle filler pattern	3.7.8.6	3.7.8.6 (1)										26			
Prime power interface	3.7.9	4.7.9 (D)										27			
Interchangeability	3.11	4.11										28			
6. Group D Inspection															
Reliability	3.10	4.10												8 Units	

(A) Analog Applique CAAs in card slots A6 through A13

(D) Loop Modem CCAs in card slots A6 through A13

(1) One time only engineering test to be covered by a seperate engineering report not part of the first article report.

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4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

	Where Required
Printed wiring	3.2.2
Semiconductor devices and intergrated circuits	3.2.3
Parts, materials and processes	3.3
Finish	3.4
Marking	3.5

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4. and 4.5.1 through 4.5.5. This does not relieve the contractor of the responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order and each subassembly and assembly identified in table II shall be inspected for conformance to the requirements as specified in table II.

4.5.1.1 Order of inspection within Group A. Group A inspections shall be performed in the order listed in table II. The order of inspections of the subassemblies, assemblies, and units (end equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the assemblies and subassemblies identified in table II shall be subjected to a 100% visual/mechanical and electrical inspection by the contractor for conformance with the inspection requirement provided in table II.

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b. Only for spares that are not installed into end equipment, the following shall be done: After completion of the above 100% inspection by the contractor, discrete lots shall be formed from the subassemblies and assemblies which passed the 100% inspection and these shall be subjected to sampling inspection utilizing the procedures of MIL-STD-105 using general inspection levels and AQLs indicated in table II (Quality Assurance Verification). Factors of lot composition not defined herein, or in the contract or purchase order shall be in accordance with MIL-STD-105.

c. Assemblies and subassemblies which have passed the above 100% visual/mechanical and electrical inspections of step a above shall be assembled into the end equipments, which shall then be subjected by the contractor to the burn-in test requirements of 4.6 and a 100% end equipment visual/mechanical and electrical inspection in accordance with table II, in that order.

d. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from units which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using the general inspection levels and AQLs indicated in table II. Group A electrical inspection shall be performed in the order shown in table II.

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TABLE II. Group A inspection.

Inspection	Reqt Para	Insp Para	AQL Major	AQL Minor
Visual-Mechanical (Including Spare Assemblies)				
BITE 2	3.14, 3.15	4.13	2.5%	4.0%
Timing generator	3.14, 3,15	4.13	2.5%	4.0%
Loop modem	3.14, 3.15	4.13	2.5%	4.0%
Mux/demux	3.14, 3.15	4.13	2.5%	4.0%
Group framing	3.14, 3.15	4.13	2.5%	4.0%
Interface	3.14, 3.15	4.13	2.5%	4.0%
Multi Vdc	3.14, 3.15	4.13	2.5%	4.0%
52 Vdc	3.14, 3.15	4.13	2.5%	4.0%
AC input	3.14, 3.15	4.13	2.5%	4.0%
Analog applique unit	3.14 3.15	4.13	2.5%	4.0%
Case and cover	3.14, 3.15	4.13	2.5%	4.0%
Electrical Performance (Including Spare Assemblies)				
BITE 2	3.12	4.12	2.5%	*
Timing generator	3.12	4.12	2.5%	*
Loop modem	3.12	4.12	2.5%	*
Mux/demux	3.12	4.12	2.5%	*
Group framing	3.12	4.12	2.5%	*
Interface	3.12	4.12	2.5%	*
Multi Vdc	3.12	4.12	2.5%	*
52 Vdc	3.12	4.12	2.5%	*
AC input	3.12	4.12	2.5%	*
Analog applique unit	3.12	4.12	2.5%	*
Electrical (unit level)				
Idle channel noise	3.7.5.8.1	4.7.5.8.1(M)	1.0%	*
BITE and alarms	3.7.10	4.7.10 (M)	1.0%	*
Digital channel check	3.7.11.1	4.7.11.1 (M)	1.0%	*
Alarm monitor check	3.7.11.2	4.7.11.2 (M)	1.0%	*
Sync frame check	3.7.11.3	4.7.11.3 (M)	1.0%	*
DC voltage check	3.7.11.4	4.7.11.4 (M)	1.0%	*
Analog channel check	3.7.11.5	4.7.11.5 (M)	1.0%	*

*All electrical defects are major

(M) Analog Applique CCAs in card slots A6 through A9, or A10 through A13 and Loop Modem CCAs in card slots A10 through A13 or A6 through A9.

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4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to table III and to the special inspection levels of table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B AQL shall be as shown in table III and the inspection level shall be S-4.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in the order listed in table III.

TABLE III. Group B inspections.

Inspection	Reqt Para	Insp Para	AQL
Line-to-line voltage levels	3.7.1.1	4.7.1.1	4.0%
Receiver dynamic range	3.7.1.7	4.7.1.7	4.0%
Output voltage	3.7.2.1	4.7.2.1 (D)	4.0%
Input voltage	3.7.2.4	4.7.2.4 (D)	4.0%
TED resync command voltage	3.7.6	4.7.6	4.0%

(D) As in Table I.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in table IV. Samples shall be selected in accordance with 4.5.3.1. An additional unit is required to perform operational checks in environmental testing.

4.5.3.1 Sampling for Group C inspections.

4.5.3.1.1 Subgroup 1. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection one unit shall be selected every third month. The unit shall be subjected to high and low temperature testing in that order.

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4.5.3.1.2 Subgroup 2. For this subgroup, three units shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fourth month, and two units shall be selected every eighth month. The unit selected every fourth month shall be subjected to vibration and shock testing, in that order. Of the units selected every eighth month, one unit shall be subjected to EMI testing and the other unit subjected to TEMPEST testing.

4.5.3.1.3 Subgroup 3. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fifth month. The unit shall be subjected to humidity testing.

4.5.3.1.4 Subgroup 4. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspections, two units shall be selected every twelfth month. One unit shall be subjected to altitude, dust and salt fog testing and one unit shall be subjected to fungus testing.

4.5.3.1.5 Subgroup 5. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fourth month. The unit shall be subjected to the electrical tests in the order shown in table IV.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in the order specified in 4.5.3.1.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

TABLE IV. Group C inspections.

Inspection	Reqt Para	Insp Para
Subgroup 1		
High temperature	3.8.1.1	4.8.1.1
Low temperature	3.8.1.2	4.8.1.2

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TABLE IV. Group C inspection (contd).

Inspection	Reqt Para	Insp Para
Subgroup 2		
EMI	3.9	4.9
Vibration	3.8.7	4.8.7.1
Shock	3.8.7	4.8.7.2
TEMPEST	3.17	4.17
Subgroup 3.		
Humidity	3.8.2	4.8.2 (Note 1)
Subgroup 4.		
Altitude	3.8.3	4.8.3
Dust	3.8.4	4.8.4 (Note 1)
Salt fog	3.8.5	4.8.5 (Note 1)
Fungus	3.8.6	4.8.6 (Note 1)
Subgroup 5.		
Clock/data phasing	3.7.1.3	4.7.1.3
Signal balance	3.7.1.5	4.7.1.5
Slew rate	3.7.1.6	4.7.1.6
Input load impedance	3.7.1.8	4.7.1.8
Clock asymmetry	3.7.1.9	4.7.1.9
Timing modes	3.7.1.10	4.7.1.10
Output waveshape	3.7.2.2	4.7.2.2 (D)
Input/output impedance	3.7.2.3	4.7.2.3 (D)
Loop power feed (surge current only)	3.7.3	4.7.3 (D)
Analog channel input/ output impedance	3.7.5.1	4.7.5.1 (A)
Longitudinal balance	3.7.5.2	4.7.5.2 (A)
TED resync command waveshape	3.7.6.1	4.7.6.1
Frame squelch	3.7.7	4.7.7
Frame acquisition time	3.7.8.2	4.7.8.2

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TABLE IV. Group C inspections (contd).

Inspection	Reqt Para	Insp Para
Detection of out-of-sync	3.7.8.4	4.7.8.4
Delayed frame search	3.7.8.5	4.7.8.5
Prime power interface	3.7.9	4.7.9 (D)
Interchangeability	3.11	4.11

(A), (D) As in table I.

NOTE 1: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is reworked to specification requirements and the sample units are resubjected to and pass Group A inspection only.

4.5.5 Group D inspection. This inspection shall consist of the tests specified in table V and shall be performed on units from lots which have been subjected to and have met Group A and Group B inspection. Test conditions, methods, and procedures shall be in accordance with 4.10.

TABLE V. Group D inspection.

Inspection	Requirement Paragraph	Inspection Paragraph
Reliability	3.10	4.10

4.5.5.1 Sampling for Group D inspection. Eight units shall be selected at random from the first month's production and subjected to the test of 4.10. For subsequent Group D inspections, a maximum of eight units shall be selected at random from every fourth month's production.

4.5.5.2 Group D failures. Actions required relative to Group D failures shall be as specified in the contract or purchase order.

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4.5.5.3 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and have passed Group D inspection may be accepted on the contract provided all damage is reworked to specification requirements and the sample units are resubjected to and pass Group A inspection only.

4.6 Electrical preconditioning (burn-in). (See 3.6) During electrical preconditioning, the equipment shall be connected as shown in figures 2 and 3. Test conditions of 4.10.3 shall apply except that duty cycle shall be 100 percent. If a failure occurs during this preconditioning, the failure shall be recorded, the failed component(s) replaced, and the preconditioning cycle reinstituted. Failure data obtained shall be submitted with the lot containing the unit in which the failure occurred. The procedure shall be as follows:

a. Set all UUTs for 32 kb/s Channel Rates, even numbered UUTs to loop timing and odd numbered UUTs to Station Clock timing. Set Timing Source and UUT Group Rate switches to Rate 7.

b. Connect the DSVT Simulator to the Loop Modem channels 9 through 16 via the Power Feed Load Box and turn ON all Loop Modem Power Feed switches. Do not connect any AAU channels.

c. Turn power ON to all UUTs. Run the units for 48 hours, using BITE to indicate failure. If a failure is detected, it is to be repaired and recorded. The last 24 hours of burn-in must be failure free.

d. After completion of burn-in, perform the operational test procedure outlined in 4.7.11.

e. The S,SA alarms shall be latched to detect any alarm occurring during the 48-hour burn-in period.

4.7 Electrical tests. (See 3.7) Unless otherwise specified, all tests shall be performed at 115 V ±5% from a single-phase source operating between 57.0 and 63.0 Hz. Tests shall be conducted at room ambient temperature, pressure, and humidity per MIL-STD-810 unless otherwise specified. The electrical test equipments listed in table VI or their equivalents will be utilized during testing and will be calibrated in accordance with MIL-STD-45662. The standard test equipment shall be provided by the contractor. Government technical representative approval is required prior to use of proposed substitute items. Unless otherwise specified, special test equipment (see table VII and figure 4) shall be provided by the contractor subject to

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approval by the Government. An alternate item of Government furnished equipment (GFE) is shown in table VIII. Table IX identifies the external interface connections with the test figures. Unless otherwise noted all resistors in test figures shall be .5 watts.

TABLE VI. Standard test equipment.

Description	Mfg/Model No. (or equivalent)
Electronic Counter	HP 5248L
Universal Counter	HP 5315A
Volt-Ohm Meter	Simpson 260
Wattmeter	Weston 432
Attenuator	Kay 432
Storage Oscilloscope	Tektronix 7613
Oscilloscope	Tektronix 7904
AC Voltmeter	HP 403B
Oscillator	HP 200CD
AC Power Source	Elgar 401-V
Transmission Measuring Set	HP 3555B
Envelope Delay Test Set	HP 4942A
Distortion Meter	HP 339A
Pulse Generator	TEKTRONIX PG 505
DC Power Source	SORENSEN QRD 30-1
Multimeter	Fluke 8050A
Frequency Synthesizer	HP 3325A

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TABLE VII. Special test equipment.

Equipment Name	Drawing No.	Functional Description
Group Simulator	A3051667	The Group Simulator simulates the group data, and subchannel orderwire and telemetry signals. It generates either a long or short pseudo-random sequence. The three signals can be generated simultaneously or in any combination.
Timing Source	A3051666	The Timing Source provides four independent group outputs, switch-selectable to positions between 72 and 4608 kb/s. Also provided are four independent Station Clock/ Telemetry Clock outputs, switch-selectable to 2, 16, or 32 kHz.
Timing Source (Alternate)		An alternate Timing Source is shown in figure 4 that may be used in lieu of the above.
DSVT Simulator	A3051668	The DSVT Simulator provides a pseudo-random pattern generator and a correlator with a conditioned diphase interface at 16 and 32 kb/s. An error output is provided for counting errors externally.
Error Inserter	A3051665	The Error Inserter provides a means of injecting a .1% continuous BER with 20% random bursts.

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Table VIII. Government furnished equipment.

Equipment	Description
Alternate (for DSVT Simulator): Generator, Digital Data SG-1139/G	As an alternate for the DSVT Simulator, the Digital Data Generator provides pseudo-random data patterns of 72 to 4,608 kb/s balanced signals and 16/32 kb/s conditioned diphase signals. It provides correlation of digital data to perform bit-error measurements.

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TABLE IX. TD-1235 interface connections

Connector	Function	PIN LO	PIN HI	PIN Gnd	Signal Type
CHANNELS 1-8	Data In Ch 1	W	X	p	DIPHASE
	Data Out Ch 1	B	C		DIPHASE
	Data In Ch 2	AA	BB	GG	DIPHASE
	Data Out Ch 2	q	z		DIPHASE
	Data IN Ch 3	Z	r	CC	DIPHASE
	Data Out Ch 3	D	Y		DIPHASE
	Data In Ch 4	i	j	y	DIPHASE
	Data Out Ch 4	N	P		DIPHASE
	Data In Ch 5	EE	FF	HH	DIPHASE
	Data Out Ch 5	s	DD		DIPHASE
	Data In Ch 6	a	b	t	DIPHASE
	Data Out Ch 6	E	F		DIPHASE
	Data In Ch 7	h	w	x	DIPHASE
	Data Out Ch 7	M	g		DIPHASE
	Data In Ch 8	f	u	v	DIPHASE
	Data Out Ch 8	c	d		DIPHASE
CHANNELS 9-16	Data In Ch 9	W	X	p	DIPHASE
	Data Out Ch 9	B	C		DIPHASE
	Data In Ch 10	AA	BB	GG	DIPHASE
	Data Out Ch 10	q	z		DIPHASE
	Data In Ch 11	Z	r		DIPHASE
	Data Out Ch 11	D	Y		DIPHASE
	Data In Ch 12	i	j	y	DIPHASE
	Data Out Ch 12	N	P		DIPHASE
	Data In Ch 13	EE	FF	HH	DIPHASE
	Data Out Ch 13	s	DD		DIPHASE
	Data In Ch 14	a	b	t	DIPHASE
	Data Out Ch 14	E	F		DIPHASE
	Data In Ch 15	h	w	x	DIPHASE
	Data Out Ch 15	M	g		DIPHASE
	Data In Ch 16	f	u	v	DIPHASE
	Data Out Ch 16	c	d		DIPHASE

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Table IX. TD-1235 interface connections (contd)

Connectors	Function	Pin LO	Pin HI	Pin Gnd	Signal Type
GROUP	Group Data In	A	B	M	NRZ
	Group Tim In	C	D	N	SQ WV
	Group Data Out	E	F	P	NRZ
	Group Tim Out	G	H	R	SQ WV
	Station Clock In	J	K	S	SQ WV
TED & ALARM	Resync Command	A	B	K	10 us minimum
	Frame Squelch Signal	C		K	TTL
	Frame Squelch Return	D		K	
	Alarm S	E	F	K	Alm=Close Contact
	Alarm SA	G	H	K	Alm=Close Contact
POWER	Power	C	A	B	

4.7.1 Balanced equipment-side signal characteristics. (See 3.7.1)

4.7.1.1 Line-to-line voltage. (See 3.7.1.1)

a. Connect the equipment as shown in figure 5.

b. Set the UUT as follows:

Group Rate	7
Channel Rate	32 kb/s
Timing	STA CLK
Power Feed Switches	OFF

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c. Set the Group Simulator as follows:

Group Rate	7
Timing Mode	EXT
Transmit Format	MSF
Type	1
Channel Rate	32 kb/s
Data Select	RANDOM

d. Measure the Group Data and Timing Out signals. Measure the positive and negative voltage levels line-to-line.

e. Verify that the voltage levels are within the following required limits for each signal:

$\pm$ 100 mV min and $\pm$ 4.0 V max

4.7.1.2 Line-to-ground voltage. (See 3.7.1.2) Measure the line-to-ground voltage for each line using test configuration in figure 5. Verify that the requirements of 3.7.1.2 are met.

4.7.1.3 Clock/data phasing. (See 3.7.1.3)

a. Connect the equipment as shown in figure 6.

b. Set the UUT switch setting as follows:

Group Rate	7
Channel Rate	32 kb/s
Timing	STA CLK
Power Feed Switch	OFF

c. Set the Group Simulator settings as follows:

Group Rate	7
Timing Mode	EXT
Transmit Format	MSF
Type	1
Channel Rate	32 kb/s
Data Select	RANDOM

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d. Measure the delay between the zero crossings on both the data rise and fall slopes to the zero crossings of the timing rise slopes as indicated in figure 7.

e. Verify that the delay (D) is within ±240 ns for both rise and fall.

4.7.1.4 Offset. (See 3.7.1.4)

a. Connect the equipment as shown in figure 8.

b. Measure the DC offset voltage between the high and low pins of the Group Data and Timing Out signals.

c. Verify that the DC offset voltages are not greater than ±3 volts.

4.7.1.5 Signal balance. (See 3.7.1.5)

a. Connect the equipment as shown in figure 5.

b. Set the UUT as follows:

Group Rate	7
Channel Rate	32 kb/s
Timing	STA CLK
Power Feed Switches	OFF

c. Set the Group Simulator as follows:

Group Rate	7
Timing Mode	EXT
Transmit Format	MSF
Type	1
Channel Rate	32 kb/s
Data Select	RANDOM

d. Measure the Group Data and Timing Out signals. Measure the positive and negative voltage levels line-to-line.

e. Compare the positive and negative values. Determine the larger of the two values.

f. Compare the difference between the positive and negative values to 10% of the larger value.

g. Verify that the difference is less than 10% of the larger value.

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4.7.1.6 Slew rate. (See 3.7.1.6).

a. Connect the equipment as shown in figure 9. The UUT and Group Simulator settings are the same as in 4.7.1.3.

NOTE: All measurements shall be made through 100 feet of RG-108 cable.

b. With the UUT at Group Rate 7, measure the time between the +40 mV and -40 mV points around the zero voltage level of the rise and fall slopes of the Group Data and Timing Out signals as shown in figure 10.

c. Verify that the time measured in step b is not more than 10 ns for each signal.

4.7.1.7 Receiver dynamic range. (See 3.7.1.7)

a. Connect the equipment as shown in figure 11. A variable attenuator will be connected in the normal signal path during the conduct of test.

b. Set the UUT for Group Rate 7 operation. Connect the attenuators in the Group Data, Timing and Station Clock signal paths. Using the oscilloscope, adjust each attenuator to give a $\pm$50 mV peak signal referenced to ground.

c. Verify that no errors occur for 30 seconds and no UUT alarms are indicated.

4.7.1.8 Input load impedance. (see 3.7.1.8)

a. Connect the equipment as shown in figure 12.

b. Connect the test setup in sequence to the Group Data and Timing and Station Clock inputs.

c. At each signal input, adjust the audio oscillator at a frequency of 1.0 kHz for an output voltage of 1.0 volt as measured by the V1 voltmeter. Read and record the V2 voltage.

d. Calculate the input impedance of each input using the formula:

$$R = 100 \frac{V2}{V1} \text{ ohms}$$

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e. Verify that the input impedance is 100 ohms ±10% in each case.

4.7.1.9 Clock asymmetry. (See 3.7.1.9)

a. The equipment test setup is the same as in 4.7.1.3, figure 6.

b. Measure the Group Timing Out at Group Rate 7. Measure the time (A) between the rise and fall slopes at the zero voltage points. Then measure the time (B) between two successive rise slopes at the zero voltage points as indicated in figure 13.

c. Verify that A =1/2 B within ± 120 ns.

4.7.1.10 Timing modes. (See 3.7.1.10)

a. Connect the equipment as shown in figure 14. Set the UUT and Group Simulator switches as in 4.7.1.3.

b. Connect the Timing Source and the oscilloscope to Station Clock input.

c. Measure the delay between the zero crossing of the negative to positive transition of the Timing Source signal and the zero crossing of any data transition as indicated in figure 7.

d. Remove the Timing Source from the Station Clock input and set the UUT Timing switch to LOOPED. Connect the oscilloscope to Group Timing In.

e. Repeat step c.

f. Verify that the time delays (D) are within ±434 ns at group rate 7.

NOTE: A negative delay indicates that the data leads the timing. Cable length shall be as short as possible.

g. Reconnect the Timing Source to the UUT Station Clock input and set the UUT Timimg to STA CLK.

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h. Set the frequency synthesizer to 576.028 kb/s ±1 ppm and Timing Source to rate 15.

i. Verify that the UUT Station Clock input and Group Data output are in sync.

j. Set the frequency synthesizer to 575.972 kb/s ±1 ppm and repeat step i.

k. Remove Station Clock input and Group Simulator connections.

l. Measure the Group Timing output frequency and verify that it is 576 kb/s ±40 ppm.

4.7.2 Digital channel characteristics. (See 3.7.2)

4.7.2.1 Output voltage. (See 3.7.2.1)

a. Connect the equipment as shown in figure 15.

b. Set the DSVT Simulator to 32 kb/s rate, all zeroes. Set the UUT to a 32 kb/s Channel Rate, Group Rate 7, Station Clock Timing and all Power Feeds set to OFF.

c. Connect the DSVT Simulator to the first Digital Channel input. Measure the peak-to-peak channel output voltage as shown in figure 16.

d. Repeat step c for Digital Channels 2 through 16.

e. Verify that the peak-to-peak output voltage into 125 ohms is 3 volts ± 15% in all cases.

4.7.2.2 Output waveshape. (See 3.7.2.2)

a. Connect the equipment as shown in figure 15.

b. Set the DSVT Simulator to 16 kb/s rate, all zeroes. Set the UUT to 16 kb/s Channel Rate, Group Rate 7, and Station Clock Timing and all Power Feeds set to OFF.

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c. Connect the DSVT Simulator to the first Digital Channel input. Measure the channel output rise and fall times and droop. Refer to Figure 16 for a definition of these characteristics.

d. Repeat steps c and d for Digital Channels 2 through 16.

e. Verify that the rise and fall times (between the 10% and 90% points) are between 0.65 and 3.0 us and the droop is less than 5% of the peak-to-peak amplitude.

4.7.2.3 Input and output impedance. (See 3.7.2.3)

a. Connect the equipment as shown in figure 17.

b. Set the UUT for 32 kb/s Channel Rate, Group Rate 7, Station Clock Timing, and all Power Feeds set to OFF.

c. With the input setup connected to the first Digital Channel, adjust the oscillator for 125 mV at a frequency of 8 kHz (V1). Then measure the V2 voltage.

d. Calculate the input impedance using:

$$Z(in) = 125 \frac{V2}{V1} \text{ ohms}$$

e. Repeat steps c and d with the oscillator frequency set to 16, 32 and 56 kHz.

f. Repeat Steps c through e for Digital Channels 2 through 16.

g. Verify that the input requirements are met, as follows:

Frequency, kHz	Impedance, ohms
8	100-144
16	100-144
32	100-144
56	75-144

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h. With the output setup connected to the first Digital Channel Output, measure the output voltage V1 with the switch open, then measure the V2 voltage with the switch closed.

i. Calculate the impedance using:

$$Z(out) = 125 \left(\frac{V1}{V2} - 1\right) \text{ ohms}$$

j. Repeat steps h and i for Digital Channels 2 through 16.

k. Verify that the output requirements are met. (Same as for step g).

4.7.2.4 Input voltage. (See 3.7.2.4)

a. Connect the equipment as shown in figures 18 and 19.

b. Set the DSVT Simulator for 32 kb/s operation and adjust the Line Simulator (switch to LOW) for 0.150 V peak-to-peak signal at the first Digital Channel RCV terminals. Set UUT for 32 kb/s Channel Rate, Group Rate 7, Station Clock Timing, and Loop Power Feeds OFF.

c. Monitor the DSVT Simulator Error Detector for a minimum of 30 seconds. Verify that there are no errors generated.

d. Repeat steps b and c for Digital Channels 2 through 16.

e. Set the Line Simulator switch to HIGH and adjust for 4.0 V peak-to-peak signal at the first Digital Channel RCV terminals.

f. Monitor the DSVT Simulator Error Detector for a minimum of 30 seconds. Verify that there are no errors generated.

g. Repeat steps e and f for Digital Channels 2 through 16.

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4.7.3 Loop power feed. (See 3.7.3)

a. Connect the Loop Power Feed Load Circuit to the first Digital Channel as shown in figure 20.

b Set S1 to B and enable the first Digital Channel Power Feed switch on the UUT. Then measure and record the current.

c. Set S1 to A and measure the voltage. Verify that the voltage is less than 56 V.

d. Set S1 to B and momentarily short either of the data output lines to either data input line. Then short each data output and data input momentarily to ground.

e. Verify that the current returns to at least 68.8 mA.

f. Turn off the Power Feed switch on the UUT.

g. Connect the loop power surge test setup a shown in figure 21 to the first Digital Channel.

h. Turn ON the Power Feed switch and adjust the pulse generator pulse width to 45 $\pm$1 ms and repetition rate of 1 Hz. Adjust the voltage V2 to a minimum of 1.0 volt using an oscilloscope. Monitor and verify that the loop power feed does not shut off.

i. Repeat steps a through h for each remaining Digital Channel 2 through 16.

4.7.4 Idle loop. (See 3.7.4)

a. Connect equipment as shown in figure 22.

b. Set UUT to 32 kb/s channels, Group Rate 7, Station Clock Timing and Loop Power Feeds OFF, and close S1.

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c. Verify that the Data Channel bits are all digital ones at the Group Data Out.(Refer to figure 1 for frame format.)

d. Open the Group Data loop-back connection, S1.

e. Verify that the channel outputs 1 through 16 are diphase ones (16 kHz square wave).

f. Verify that the Data Channel bits are all digital zeros at the Group Data Out.(Refer to figure 1 for frame format.)

4.7.5 Analog channel characteristics. (See 3.7.5)

4.7.5.1 Input/output impedance and return loss. (See 3.7.5.1)

a. Connect the equipment as shown in figure 23. Do not connect the setup to the UUT.

b. Set the oscillator to 300 Hz and adjust the output for a V1 voltage of 0 dBm. Record the V2 (set) voltage in dBm (Z test= ∞).

c. Connect the setup to the first Analog Channel input, reset the V1 voltage to 0 dB and record the V2 (measured) voltage in dBm.

d. Repeat steps a through c for each remaining analog channel input.

e. Repeat steps a through d with the oscillator frequency set to 3400 Hz.

f. Repeat steps a through e measuring each Analog Channel output.

g. Calculate the return loss, V2 (set) minus V2 (measured) in dB and verify that it is not less than 20 dB.

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4.7.5.2 Longitudinal balance. (See 3.7.5.2)

a. Set up the bridge as shown in figure 24.

b. Check the bridge balance at 300 and 3400 Hz without the UUT using the formula in step h. It should be at least 60 dB.

c. Connect the bridge to the first Analog Channel input.

d. Set V1 to 1 Vrms at 300 Hz. Then measure the V2 voltage.

e. Repeat step d with the frequency set to 1000 Hz and 3400 Hz.

f. Repeat steps d and e with the bridge connected to the first Analog Channel output.

g. Repeat steps c through f for each of the remaining Analog Channels.

h. Calculate the longitudinal balance using:

$$LB = 20 \log \frac{V1}{V2}$$

i. Verify that the balance is not less than 40 dB for each measurement.

4.7.5.3 Frequency response. (See 3.7.5.3) The frequency response requirement is demonstrated as part of the individual CCA TRS testing performed on 100% of the CCAs.

4.7.5.4 Analog Channel input and output test tone. (See 3.7.5.4) The test tone level and frequency shall be demonstrated as part of the individual CCA TRS performed on 100% of the CCAs.

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4.7.5.5 Envelope delay distortion. (See 3.7.5.5)

a. Connect the equipment as shown in figure 25.

b. Set the Analog Channel and Output/Input Level switches to 0dB and the Channel Loopback switches to LOOPBACK. Set the channel Rate to 32 kb/s, Group Rate 7, Timing switch to STA CLK.

c. Measure the envelope delay distortion at 800 Hz and at 3000 Hz.(This includes the input and output filters of the UUT.)

d. Repeat steps a and b for the remaining Analog Channels.

e. Verify that the delay is less than 100 us for each measurement.

4.7.5.6 Input-output linearity. (See 3.7.5.6) The linearity requirement is demonstrated as part of the individual CCA TRS test performed on 100% of the CCAs.

4.7.5.7 Loopback switch. (See 3.7.5.7) Operation of loopback switches is demonstrated as part of 4.7.5.5 (for loop mode) and 4.7.5.8.1 (for normal mode).

4.7.5.8 Noise. (See 3.7.5.8)

4.7.5.8.1 Idle channel noise. (See 3.7.5.8.1)

a. The equipment connections are as shown in figure 26 with 600 ohms connected to the analog input terminals.

b. Set the Analog Channel Input and Output Level switches to 0 dB and the Channel Loopback switches to NORMAL. Set the Channel Rate to 32 kb/s, Group Rate 7, Timing switch to Station Clock.

c. Set the noise measuring set to 600 ohms terminated and C-message weighting.

d. Measure the channel output noise.

e. Repeat steps a through d for each remaining Analog Channel.

f. Verify that the noise measured does not exceed 33 dBrnC.

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4.7.5.8.2 Crosstalk. (See 3.7.5.8.2) For this test, the UUT must contain all Analog Channels.

a. Connect the equipment as shown in figure 27.

b. Set the UUT Input Level switches to -10 dB, the Output Level switches to 0 dB, Channel Loop Back switches to NORMAL, Channel Rate 32 kHz, Group Rate 7, and Station Clock Timing.

c. With the oscillator connected to Channel 1 adjust the oscillator at 820 Hz to produce an output voltage V1 of +6 dBm. (This gives a channel input of 0 dBm at 820 Hz).

d. Set the transmission measuring set to 600 ohms termination, C-message weighted. Measure the voltage at Channel 2 Out.

e. Repeat steps c and d for channel outputs 3 thru 16.

f. Verify that the voltages do not exceed 34 dBrnC.

g. Connect the equipment as shown in figure 28. Set the Auxiliary TD-1235 to Loop Timing and the UUT to Station Clock Timing.

h. With the oscillator connected to Analog Channel 1 In, adjust the oscillator at 820 Hz to produce an output voltage V1 of +6 dBm. (This gives a channel input of 0 dBm at 820 Hz).

i. Set the transmission measuring set to 600 ohms termination, C-message weighted. Measure the voltage at Channel 1 Out. Verify that the voltage does not exceed 34 dBrnC.

4.7.5.9 Peak clipping. (See 3.7.5.9)

a. Connect the equipment as shown in figure 29.

b. Set the Analog Channel Input Level switch to -10 dB, and the Output Level switch to +10 dB. Set the Channel Loopback switch to NORMAL, the Channel Rate switch to 32 kHz, Station Clock Timing, and Group Rate 7.

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c. With the oscillator connected to the first Analog Channel Input, set the oscillator output at 200 Hz for +10 dBm (This gives a channel input of +4 dBm).

d. Measure total distortion.

e. Repeat steps a through d for each remaining Analog Channel.

f. Verify that the total distortion is less than 10% for each channel.

4.7.5.10 Full load tone. (See 3.7.5.10) The full load tone is demonstrated as part of the individual CCA TRS tests performed on 100% of the analog CCAs.

4.7.5.11 Analog channel signal processing. (See 3.7.5.11)

4.7.5.11.1 Clocking rate. (See 3.7.5.11.1) Clocking rate is demonatrated as part of 4.7.5.11.2.

4.7.5.11.2 Dynamic range. (See 3.7.5.11.2) The dynamic range is demonstrated as part of the individual CCA TRS tests performed on 100% of the analog CCAs.

4.7.5.11.3 Output level control. (See 3.7.5.11.3) The output level control is demonstrated as part of the individual CCA TRS test performed on 100% of the analog CCAs.

4.7.5.11.4 Input level control. (See 3.7.5.11.4) The input level control is demonstrated as part of the indivudual CCA TRS tests performed on 100% of the analog CCAs.

4.7.6 TED resync command voltages. (See 3.7.6)

a. Connect the equipment as shown in figure 30.

b. Set the Group Simulator to MSF, ALL ONES position, 32 kb/s rate, Type 1, Looped Timing, and Group Rate 7.

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c. Set the UUT to 32 kb/s Channel Rate and Group Rate 7.

d. Set the oscilloscope to measure Channel 1 minus Channel 2 and verify that the resync command level is between -0.1 and -3.0 volts and no pulse is present.

e. Set the Group Simulator to NO FRAME, ALL ONES position.

f. Measure the resync command amplitude.

g. Verify that the pulse is present and is between +0.1 and +3.0 Volts.

4.7.6.1 TED resync command waveshape. (See 3.7.6.1)

a. The equipment setup is the same as in 4.7.6.

b. Set the Group Simulator to the NO FRAME ALL ONES position.

c. Measure the resync command pulse width and rise and fall times. Verify that the following requirements are met.

Parameter	Requirement
Pulse width @ 0 volts	10 us min, 100 us max
Rise time, 10-90%	3 us max
Fall time, 10-90%	3 us max

4.7.7 Frame squelch. (See 3.7.7)

a. Connect the equipment as shown in figure 31.

b. The UUT and Group Simulator settings are the same as in 4.7.6.1.

c. Verify that the signal appearing at I/O Pin 29 of the Interface CCA is a TTL Logic 1.

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d. Connect the oscillator as shown. Set the oscillator for 1.0 Vrms at 1 kHz as measured by the V2 voltmeter. Then measure the V1 voltage. Calculate the impedance using

$$Z = 383 \frac{1}{\frac{V1}{V2} - 1} \text{ ohms}$$

e. Disconnect the oscillator and connect the DC power supply with S1 open. Measure the voltage (V_3) across the Alarm Squelch line. Set the power supply to 5V and close S1.

f. Decrease V_3 power supply voltage to 2.0 V. Verify that the signal appearing on the oscilloscope remains a Logic 1.

g. Decrease the V_3 power supply slowly until the signal changes to a Logic 0. This is the lowest threshold level of the squelch input.

h. Verify that the following requirements are met.

Step	Requirement
d.	Impedance: 1 kilohm minimum 51 kilohm maximum
e.	V_3 between 4.5 and 5.5 V
f.	V_3 = 2.0 V minimum, Logic 1
g.	V_3 = 0.8 V maximum, Logic 0

4.7.8 Framing. (See 3.7.8)

4.7.8.1 Format. (See 3.7.8.1) The TD-1235 Output Group Patterns and Overhead Channels will be verified by examination of the Group Output Data displayed on the oscilloscope. Channel positions will be verified by sequentially applying a diphase all zero pattern and observing that the corresponding bit position in the group signal responds to the zeroes. The telemetry subchannel bit positions will be verified by applying random data from the Group Simulator to the seventh overhead subchannel (the framing subchannel counts as zero).

For this test configure the UUT with Digital Channels.

a. Connect the equipment as shown in figure 32.

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b. Set the UUT for 32 kb/s Channel Rate, Group Rate 4, all Power Feeds OFF, and Station Clock Timing.

c. Set the Group Simulator for:

Group Rate	4
Timing Mode	EXT
Transmit Format	MSF
Channel Rate	32 kb/s
Type	1
Data Select	ALL ONES

d. Observe the transmit data displayed on the oscilloscope and locate the Framing Bit. Verify that the Group Patterns and Overhead Channels are as shown in figure 33. (7 channels)

e. Repeat step d with the UUT and Group Simulator configured for Group Rates of 5, 6 and 7 (8, 15 and 16 channels).

f. Repeat step d and e with the UUT and Group Simulator Channel Rate set to 16 kb/s, and Group Rates 2, 3 4 and 5.

g. Apply a random signal to Subchannel 7 with the Group Simulator. Verify the return of the random signal from the UUT in Subchannel 7. Verify that Subchannels 1 through 6 are digital ones.

4.7.8.2 Frame acquisition time. (See 3.7.8.2)

a. Connect the equipment as shown in figure 34. Configure both of the TD-1235s for 32 kb/s Channel Rate and Group Rate 7. Set the UUT to Station Clock Timing and the Auxiliary Unit to Looped Timing.

b. Set up the error inserter for 0.1% BER continuous operation.

c. Squelch the data at the error inserter to cause a loss of frame, then release squelch to restore data.

d. Measure the time from the resumption of received group data on error inserter, to the resumption of data, I/O 48 on the group framing card. Use the time interval counter, take 50 measurements and verify that no more than 5 of the 50 measurements exceed 50 ms.

e. Set the error inserter for 20% BER random bursts interspersed with 0.1% BER random error intervals.

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f. Repeat step d, verify that no more than 5 of 50 measurements exceed 100 ms.

4.7.8.3 Frame maintenance. (See 3.7.8.3)

a. Connect the equipment as shown in figure 34. Configure both TD-1235s for 32 kb/s Channel Rate and Group Rate 7. Set the UUT to Station Clock Timing, the Auxiliary TD-1235 for Looped Timing.

b. Connect the counter in the event count mode to TP6 on the group framing card and adjust it to trigger on the positive edge of the signal. Squelch the data; this will cause the TD-1235 to reframe. The counter should advance by one. Reset the counter.

c. Set the error inserter for 20% BER random error rate.

d. Run the unit for 12 hours, verify that no declarations of out-of-sync are generated. Repeat this step.

4.7.8.4 Detection of out-of-sync. (See 3.7.8.4)

a. The equipment setup is the same as in 4.7.8.2, Figure 34.

b. Set the error inserter to no errors.

c. Squelch the data, and measure the time from traffic interrupts to out-of-frame indication.

d. Use the Data Squelch signal at the error inserter to start the time interval measurement and use the signal at TP3 on the Group Framing Card to stop the counter. Take 50 measurements.

e. Verify that no more than 5 of 50 measurements exceed 700 ms, and that none are less than 450 ms.

4.7.8.5 Delayed frame search. (See 3.7.8.5) The delayed frame search requirement of 3.7.8.5 is demonstrated as part of the "Detection of out-of-sync" test of 4.7.8.4.

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4.7.8.6. Idle filler pattern. (See 3.7.8.6) The idle filler pattern requirements of 3.7.8.6 are demonstrated as part of the "Idle loop" tests of 4.7.4.

4.7.9 Prime power interface. (See 3.7.9)

a. Connect the equipment as shown in figure 35.

b. Configure the UUT for 32 kb/s Channel Rate, Group Rate 7 and Station Clock Timing.

c. Connect all Digital Channels installed to the Loop Power Feed Load Box, such that the DSVT input and output is looped through the Digital Channels in tandem.

d. Turn on the Digital Channel Power Feed switches.

e. For each combination of input voltage and frequency given below perform the following:

(1). Measure the input power

(2). Monitor the Digital Channel BER over a 30-second period. Verify that no errors are generated.

(3). Measure the gain on each Analog Channel installed. The gain is measured as follows:

Connect the oscillator through 600 ohms to the Analog Channel input. Adjust the oscillator at 820 Hz for -11 dBm. Then measure the Analog Channel output voltage in dBm terminated in 600 ohms.

NOTE: Gain in dB = UUT output minus actual output of oscillator plus 6 dB.

(4). Verify that the gain is 0 ±1 dB, on each Analog Channel.

VOLTAGE	FREQUENCY
115 Vac	60 Hz
103.5 Vac	47.5 Hz and 420 Hz
126.5 Vac	47.5 Hz and 420 Hz

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4.7.10 BITE and alarms. (See 3.7.10).

a. Connect the equipment as shown in figure 26.

b. Set the UUT Power Feed switches to OFF, the Channel Rate switches to 32 kb/s, Station Clock Timing, and Group Rate 7. Do not connect the Analog Channel inputs.

c. Connect the unit to 115 Vac power and turn the unit ON. Verify that the following BITE and alarm conditions are met. (Open circuit is the normal state. A short indicates alarm).

Requirement
S open
SA open
Summary Fault, Power Out of Tolerance (POT) and Loss of Input (LOI) lamps OFF. All CCA LEDs OFF.
All CCA LEDs OFF.

d. Press the Lamp Test switch on the UUT. Verify that the BITE requirements listed below are met.

Requirement
S closed
SA closed
Summary Fault, POT Lamp and LOI lamps illuminate
BITE LEDs on CCAs illuminate in sequential order during self test
All BITE indicators are OFF after self-test

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a. Turn OFF the power at the source. Do not turn the UUT power switch off. Verify that the following BITE requirements are met: S closed, SA closed.

b. Turn OFF the UUT power switch. Verify that the following BITE requirements are met: S closed, SA open.

4.7.11 Operational checks. (See 3.7.11 and 3.13)

a. The equipment connections are the same as in 4.7.10, figure 26.

b. Set all Analog Channel Loopback switches to NORMAL and Level switches to 0. Set all Loop Power Feed switches ON.

c. Set the UUT for 32 kb/s Channel Rate, Station Clock Timing, Group Rate 7, and loopback the Group signal.

d. Connect the UUT to 115 Vac power and turn the unit ON.

4.7.11.1 Digital channel check. (See 3.7.11.1)

a. Connect the DSVT Simulator to the Looped Digital Channels.

b. Monitor the Simulator error output for a 30-second period.

c. Verify that there are no errors.

4.7.11.2 Alarm monitor check. (See 3.7.11.2)

a. Monitor the S, LOI and SA alarms. Verify that the following BITE conditions are met.

Requirement
S open
SA open
Summary Fault, POT and LOI lamps OFF
All CCA LEDs OFF

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b. Open S1 on the UUT Group (do not disconnect the Station Clock input). Verify that the following BITE conditions are met.

Requirement
S Closed
SA Open
LOI lamp illuminates

4.7.11.3 Sync/frame check. (See 3.7.11.3)

a. Using the oscilloscope, monitor the resync command pulse, open S1 on UUT Group connector. Verify the presence of a resync command pulse. Close S1 on UUT Group, verify that no resync command pulse is generated and that there are no digital errors for 30 seconds indicating that the unit has reframed.

4.7.11.4 DC voltage check. (See 3.7.11.4)

a. Measure the dc voltage on each digital line at the Power Feed Load Box between input and output terminals of each channel.

b. Verify that the voltage is between 46.9 and 56 volts dc on each channel.

4.7.11.5 Analog channel check. (See 3.7.11.5)

a. Connect the analog oscillator to the first Analog Channel through 600 ohms. Set the oscillator at 820 Hz for an output of -11 dBm.

b. Connect the Analog Channel output to the voltmeter terminated in 600 ohms.

c. Measure the first Channel output gain in dBm.

NOTE: Gain in dB = UUT output minus actual output of oscillator + 6dB.

d. Repeat steps a through c for each remaining Analog Channel.

e. Verify that the gain is 0 $\pm$ 1 dB, for each channel.

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4.7.12 Environmental operational measurements. (See 3.7.12)

a. Connect the equipment as shown in figure 36.

b. Set the UUT and Auxiliary Unit for 32 kb/s Channel Rate and Group Rate 7. Set the UUT to Station Clock Timing and the Auxiliary Unit to Loop Timing.

c. Connect the DSVT Simulator to the Looped Channels on the Auxiliary Unit.

d. Connect the UUT and the Auxiliary Unit to 115 Vac power.

4.7.12.1 Digital channel check. (See 3.7.12.1) Monitor the Simulator error output for a period of 30 seonds. Verify that there are no errors.

4.7.12.2 Alarm monitor check. (See 3.7.12.2)

a. Monitor the S and SA alarm outputs at the UUT. Verify that there are no alarms generated, that is, there is an open circuit indicated on each ohmmeter. (A short indicates alarm).

b. Press the Lamp Test switch on the UUT and verify that the S and SA alarm outputs are shorted.

4.7.12.3 Sync/frame check. (See 3.7.12.3)Monitor the UUT Summary Alarm while turning OFF the Auxiliary Unit. The UUT shall generate a Summary Fault. Turn ON the power to the Auxiliary Unit. The UUT Summary Alarm shall extiguish and error activity shall cease indicating that the unit has reframed.

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4.7.12.4 DC voltage check. (See 3.7.12.4) Measure the DSVT power feed voltage at each of the 16 power feed loops on the UUT with a dc voltmeter. Verify that each output is between 46.9 and 56 Vdc.

4.7.12.5 Analog Channel check (See 3.7.12.5) In this test the analog channel gain in both directions, from Auxiliary TD-1235 to UUT and from UUT to Auxiliary TD-1235 is checked.

a. Connect the oscillator at 820 Hz terminated in 600 ohms to the first Analog Channel input on the UUT. Set the oscilltor for an output of -11 dBm.

b. Measure the first Analog Channel output on the Auxiliary TD-1235 with the voltmeter terminated in 600 ohms.

NOTE: Channel Gain in dB = UUT output minus actual oscillator output plus 6 dB.

c. Repeat steps a and b in the other direction, that is into Auxiliary Unit Channel 1 and out of UUT Channel 1.

d. Repeat steps a thru c for each remaining Analog Channel.

e. Verify that the gain is 0 dB +1 dB in any direction.

4.8 Service conditions tests. (See 3.8)

4.8.1 Temperature. (See 3.8.1)

4.8.1.1 High temperature. (See 3.8.1.1) The equipment shall be subjected to the test of MIL-STD-810D, Method 501.2 Procedure II for cycling exposure. In step 12, raise the internal chamber temperature to +63°C (+145°F). Maintain internal chamber temperature for 6 hours at +63°C (+145°F). Raise the internal chamber temperature to +71.1°C (+160°F) within a time period of one hour and maintain at that temperature for four additional hours. Lower the internal chamber temperature to +63°C (+145°F) within a time period of one hour. Repeat the above steps two additional times for a total of three 12-hours cycles. Step 14 shall be performed as follows: adjust the internal chamber temperature to +63°C (+145°F) and maintain this temperature until temperature stabilization of the test item is reached. Operate the test item until the item is stabilized.

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4.8.1.2 Low temperature. (See 3.8.1.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 502.2, Procedure I. Lower the internal chamber temperature in step 2 to -57°C(-70°F) for a period of not less than two hours following stabilization of the equipment, or until 24 hours have elapsed, whichever occurs first. In step 5 of Procedure I, perform Procedure II - Operation. (Para. II- 3.2 of Method 502.2). The chamber temperature in step 1 of Procedure II shall be adjusted to -31.7°C(-25°F).

4.8.2 Humidity. (See 3.8.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 507.2, Procedure III.

4.8.3 Low pressure (altitude). (See 3.8.3) The equipment shall be subjected to the test of MIL-STD-810D, Method 500.2, Procedure I. For step 2, adjust the chamber pressure to 141 mm of Hg (5.56 inches of Hg or 40,000 feet above sea level), at a rate not to exceed 2,500 fpm. The equipment shall also be subjected to the test of MIL-STD-810D, Method 500.2, Procedure II. The chamber pressure in step 2 shall be adjusted to 523 mm of Hg (20.6 inches of Hg or 10,000 feet above sea level) at a rate not to exceed 2,500 fpm.

4.8.4 Dust. (See 3.8.4) The equipment shall be subjected to the test of MIL-STD-810D, Method 510.2, Procedure I. In step 2 and step 5, the air velocity shall be 200 ±100 feet per minute. In step 5, the air temperature shall be adjusted to +63°(+145°F). The equipment shall be reoriented periodically to expose all its sides to the dust stream. The equipment shall be operated before step 1, during step 8 and during step 11.

4.8.5 Salt fog. (See 3.8.5) The equipment shall be subjected to the test of MIL-STD-810D, Method 509.2, Procedure I. In step 2, the 48 hour period shall be constant wetting. Delete step 3. Operation in step 4 shall take place after gentle wash in running water not warmer than +38°C(+100°F) and drying. There shall be no corrosion of finishes or metals. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered or otherwise loosened finish on metal surfaces. There shall be no clogging or binding of moving mechanical parts.

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4.8.6 Fungus. (See 3.8.6) The equipment in the assembled-and-ready-for-delivery condition, or representative samples subject to approval of the procuring agency, shall be subjected to the test of MIL-STD-810D, Method 508.3. After 28 days, based on visual examination, the equipment or samples shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10% or less of the total area and no more than six unrelated minute colonies, with mycelial development in non-critical areas. Critical areas include: terminal spacing, CCAs, etc. The equipment shall fail if it shows more than the growth specified above.

4.8.7 Vibration and shock. (See 3.8.7)

4.8.7.1 Vibration. The equipment shall be subjected to the test of MIL-STD-810D, Method 514.3, Procedure I, Category 8 (Ground Mobile). The vibration level shall be in accordance with figures 514.3-7 through 514.3-9 of MIL-STD-810D. The axes designated on the schedules are the vehicle axes. The orientation of the equipment relative to the vehicle, therefore, must be considered for proper direction of vibration input. Test time shall be 4 hours in each of three mutually perpendicular directions for a total of 12 hours. Prior to and following these tests, the equipment shall be operated in accordance with MIL-STD-810D general test performance guidance paragraph 4.5. The equipment shall not have any degradation in performance and shall not suffer any mechanical damage.

4.8.7.2 Shock.

4.8.7.2.1 Functional. The equipment shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure I. The shock pulse shall be 20g peak for a nominal duration of 11 msec in accordance with figure 516.3-4.

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4.8.7.2.2 Bench handling. The equipment shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure VI.

4.9 EMI tests. (See 3.9) The unit shall be tested for compliance with 3.10. Test methods and procedures shall be in accordance with MIL-STD-462, and a Government-approved test plan. During susceptibility testing, the integral test facility plug-in subassembly shall be used to monitor operation of the unit.

4.9.1 Bonds and grounds. (See 3.9.1) Prior to performance of EMI testing, compliance with requirements of 3.10.1 shall be ascertained. Bonding resistance measurements shall be recorded and included in the EMI test report.

4.10 Reliability testing. (See 3.10)

4.10.1 First article reliability test. First article inspection reliability testing shall be in accordance with Test Plan VIC of MIL-STD-781C.

4.10.2 Quality conformance (Group D) reliability test. The reliability test for Group D inspection shall be in accordance with Test Plan VIC of MIL-STD-781C.

4.10.3 Reliability test conditions. Reliability testing shall be accomplished subject to the following conditions:

a. Chamber temperature: 40°C ±5°C (95°F to 113°F)

b. Duty cycle shall be as follows: Apply prime power to equipment under test for 4 hours. Turn power OFF for 2 hours. Turn power ON for 16 hours. Turn power OFF for 2 hours. Repeat this sequence for each 24-hour cycle.

c. Vibration: Sinusoidal, 2.2 g ±10% peak acceleration at non-resonance frequency between 20 and 60 Hz measured at equipment mounting points. Apply for 10 minutes during each hour of testing.

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4.10.4 Reliability test method. The detailed test procedures, and conditions for performing the tests outlined in 4.10.1 and 4.10.2 shall be in accordance with contractor-prepared, Government-approved test plan.

4.10.5 Failure definition. A failure is defined as any malfunction which causes or may cause inability to commence operation, cessation of operation, or degradation below limits specified in 3.7.11.

4.10.6 Failure reporting and analysis. A failure reporting and analysis system shall be implemented per MIL-STD-781C. Final determination of the failure relevancy and multiple and pattern failure categorization shall be the responsibility of the procuring activity.

4.11 Interchangeability. (See 3.11) The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.12. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of switches, sockets, and fasteners.

d. Location of panel-mounting screw holes on basic unit for mating with case.

e. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts or plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

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4.12 Electrical subassembly tests. (See 3.12) The unit subassemblies, including spares, shall be tested in accordance with their respective TRS drawings listed in 3.12.

4.13 Visual and mechanical. (See 3.14 and 3.15) The equipment shall be examined for the defects listed in MIL-STD-252 to determine compliance with 3.15. An inspection shall be performed to verify compliance with those portions of 3.14 (System safety engineering) which can be determined visually.

4.13.1 System Safety Design Verification Check List. An inspection shall be performed, on one unit only, during first article testing, utilizing SEL Form 1183, to verify compliance with 3.14 (System safety engineering.)

4.14 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.

b. Quality conformance inspection of packaging.

4.14.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.14.2 Quality conformance inspection of packaging.

4.14.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.14.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.14.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table X, "Packing inspection provisions". Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and AQL of 4.0% defective.

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TABLE X. Packing inspection provisions.

NO.	CHARACTERISTIC	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

4.15 Serial numbers. (See 3.5.1) The equipment shall be inspected to verify that the serial numbers appear on each nomenclatured item and CCA.

4.16 Nuclear tests. (See 3.16) Applicable tests for nuclear survivability shall be performed on the equipment in accordance with Section 3 of CR-CX-0173-001.

4.17 TEMPEST. (See 3.17) TEMPEST tests shall be performed on the equipment in accordance with CR-CX-0174-001.

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5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-2073-2, tables I through VII, coded as follows:

5.1.1 Level A. 3Q-1-00-00-NR-X-ED

5.1.2 Level B. 2M-1-00-00-NR-X-ED.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The TD-1235 will be used in shelters and vans and will be mounted in an equipment case. It combines up to 16 16/32 kb/s loop circuits via WF-16 field wire into a loop group at the rates from 128 kb/s to 576 kb/s. It also provides power feed to interconnecting subscriber sets, KY-68(DSVT) and TA-954/TT (DNVT) It interfaces with modems and other multiplexers in the DGM family of equipments. An optional analog capability is also provided for interconnecting to subsets TA-341, TA-720 and TA-838.

6.1.1 Functional description. The TD-1235 is designed to time divison multiplex/demultiplex 7, 8, 15 or 16 analog or conditioned diphase loops into an 8-, 9-, 16-, or 18-channel group. The first channel of the group is allocated to overhead information including a 2/4 kb/s framing subchannel and a 2 kb/s SYSCON telemetry subchannel. The TD-1235 operates with the KY-68/94 Digital Secure Voice Terminal (DSVT) as a loop input/output via WF-16 telephone cable. The group input/output is balanced non-return-to-zero (NRZ) data and timing, and is compatible with the KG-81 Trunk Encryption Device (TED), TD-1236/G Trunk Group Multiplexer, TD-1237(P)/G Master Group Multiplexer, MD-1026(P)/G Group Modem and MD-1023(P)/G Low Speed Cable Driver Modem.

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6.2 Ordering data. Procurement documents should specifiy the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A or B preservation and packing (see Section 5).

c. When first article rough handling tests are not required.

d. Quantity of first article inspection units required.

e. Place of final inspection.

f. Technical literature required.

g. Quantity of tools and running spare parts required.

h. Group C failure requirements.

i. MIL-Q-9858.

j. Length of cables (if any are required).

k. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packing.

6.3 Environmental. Environmental pollution prevention measures are contained in the packing material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposabililty methods.

6.4 Final inspection. (See 4.5.3.3) Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.

6.4.1 Initial shipments. (See 4.4.5.3.3) Initial shipment of end items will be contingent upon successful completion of all first article tests.

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6.5 Location of operational inspection. (See 4.7.11) It is desirable that the operational system test be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in the specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.7 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage, and handling.

6.8 Government-furnished property. The contracting officer should arrange to furnish the property listed in table VIII.

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.10 Subject terms (key word) list.

Analog Applique
Balanced
Continuous Variable Slope Delta
CVSD
Digital Group Multiplex
Framing
Loop Group Multiplex (LGM)
Loop Modem
Multiplexer
Multiplexer, TD-1235(P)/TTC
MUX/DEMUX

CUSTODIAN:

ARMY - CR

PREPARING ACTIVITY:

ARMY - CR
Project No. 5820-A831

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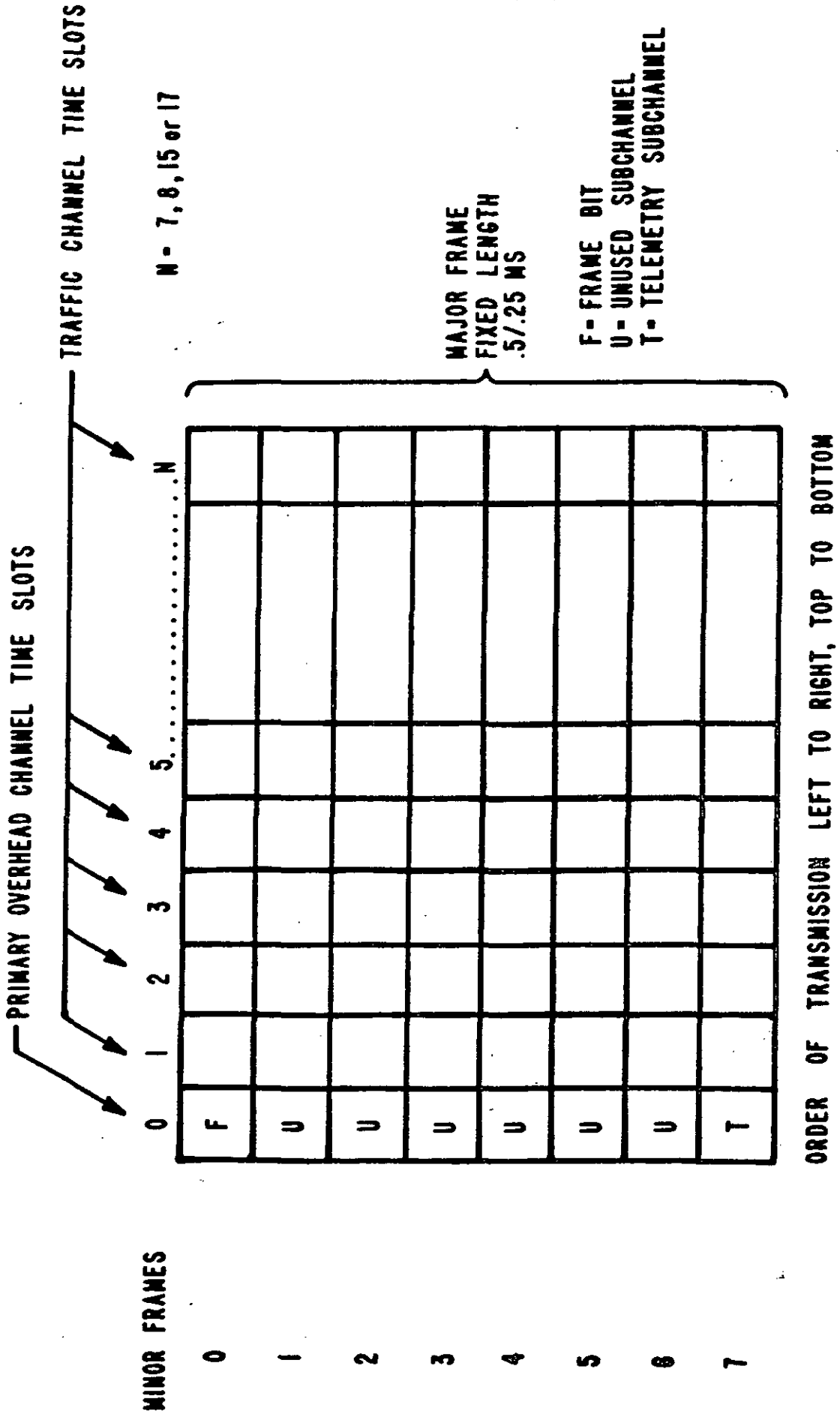


Figure 1. TD-1235 overhead channel format.

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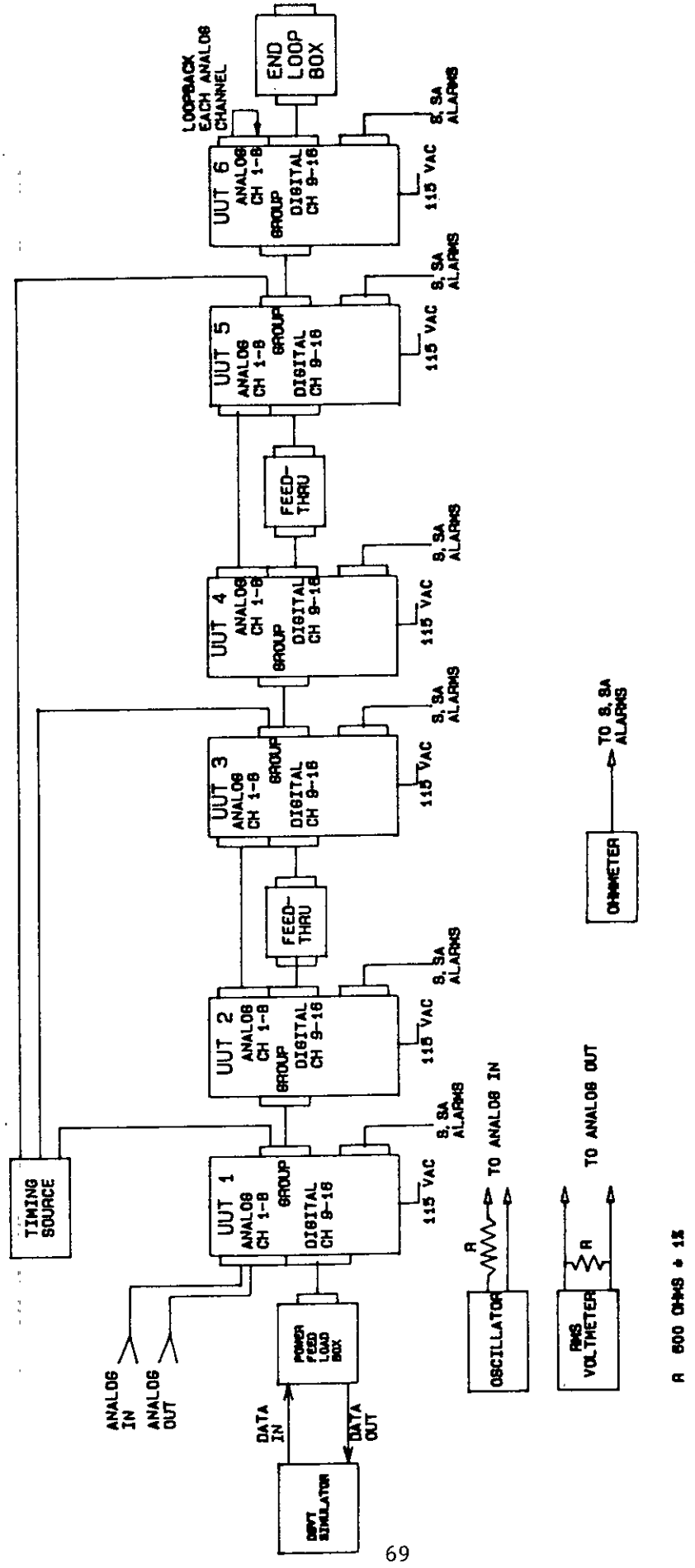


FIGURE 2. Burn-in.

For detail of test boxes see figure 3.

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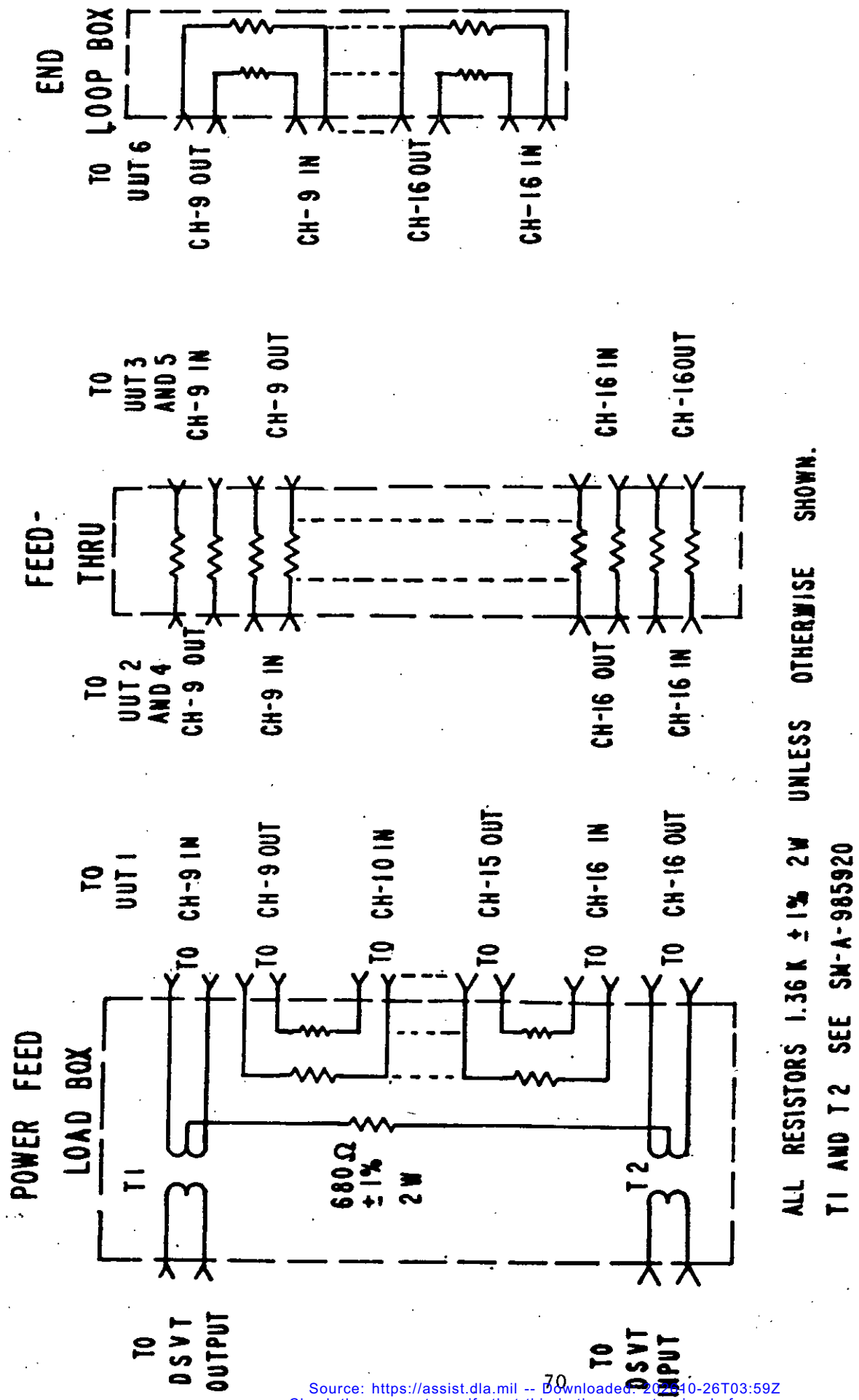
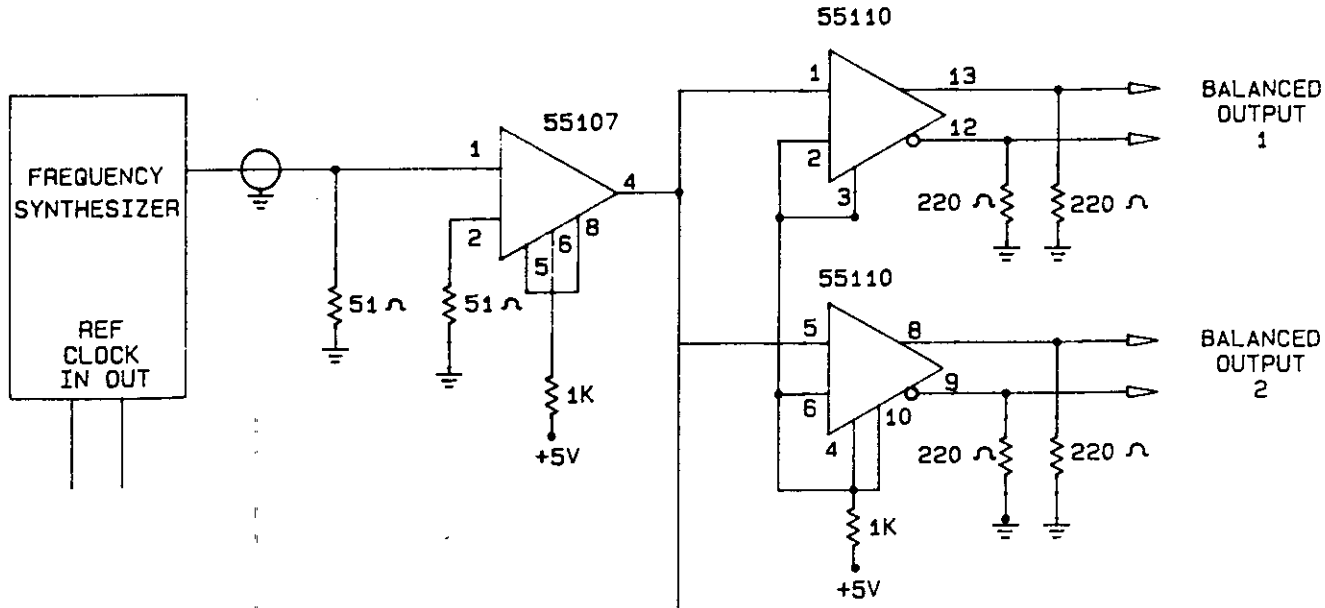


Figure 3. Test boxes (burn-in).

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GROUP RATE	SYNTHESIZER FREQUENCY
1	72 KHz
2	128 KHz
3	144 KHz
4	256 KHz
5	288 KHz
6	512 KHz
7	576 KHz
8	1024 KHz
9	1152 KHz
10	1536 KHz
11	2048 KHz
12	2304 KHz
13	4096 KHz
14	4608 KHz
15	4915.2 KHz

FIGURE 4. Alternate timing source.

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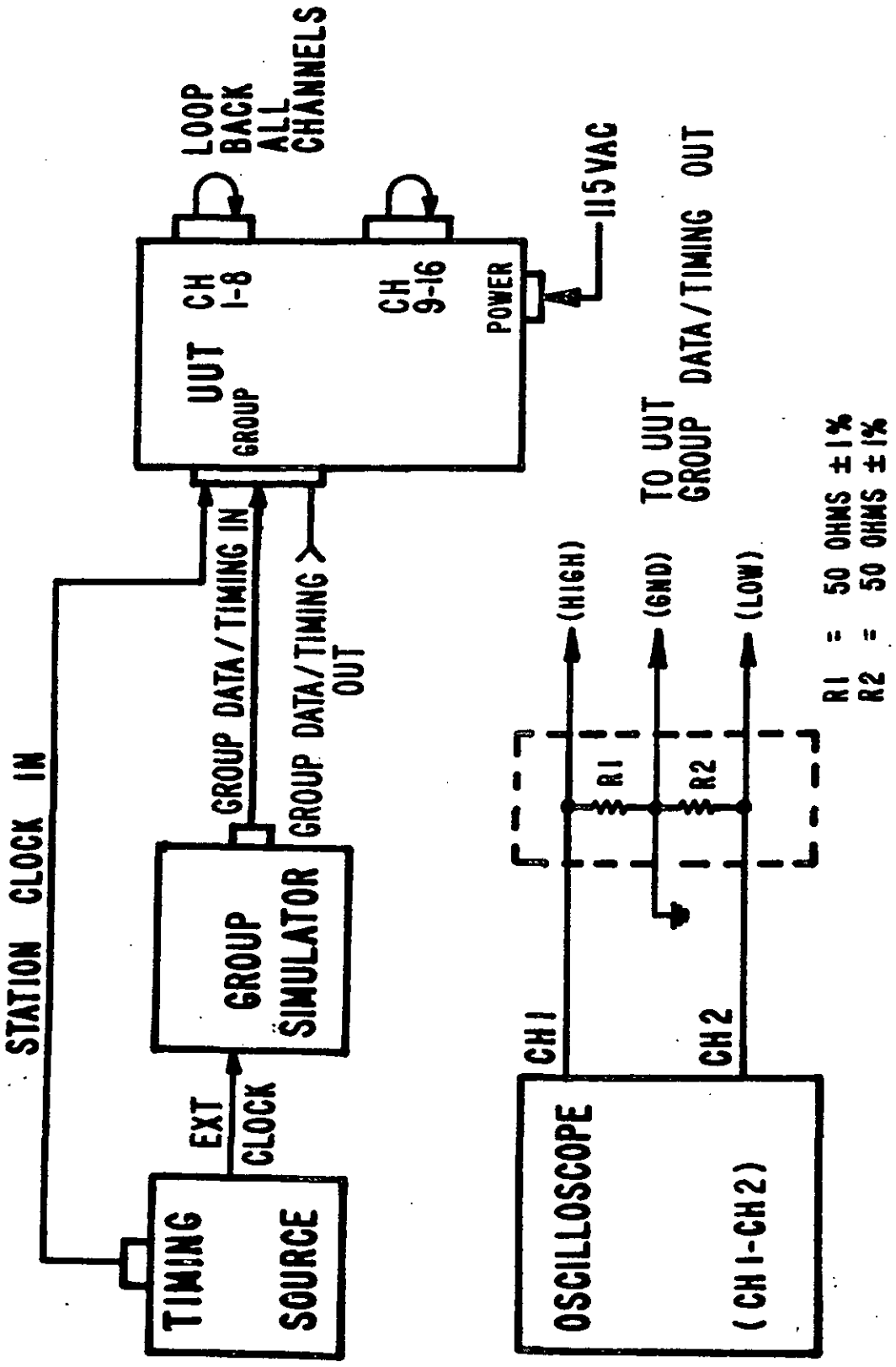


Figure 5. Line-to-line voltage levels; signal balance.

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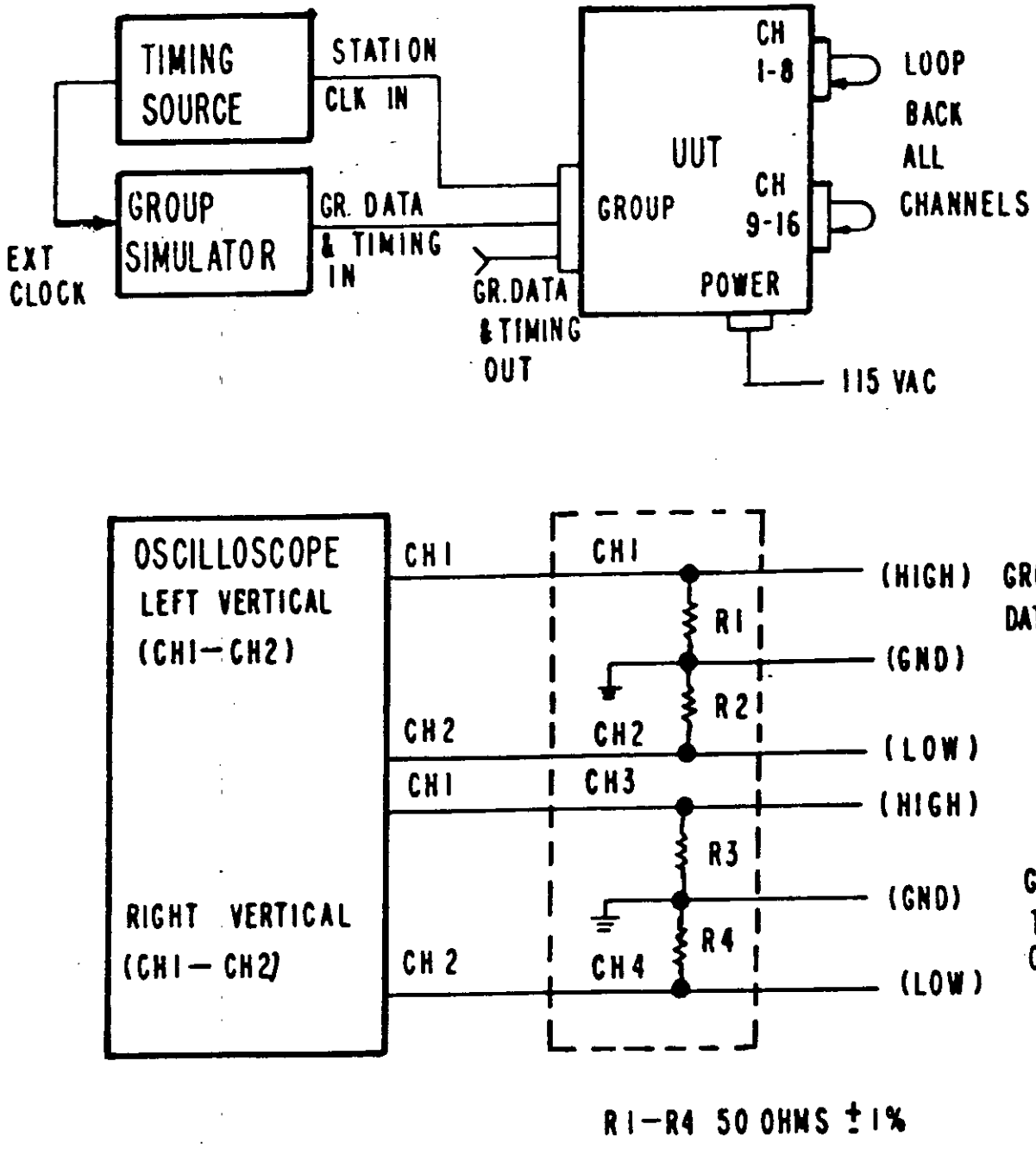


Figure 6. Clock/data phasing; asymmetry.

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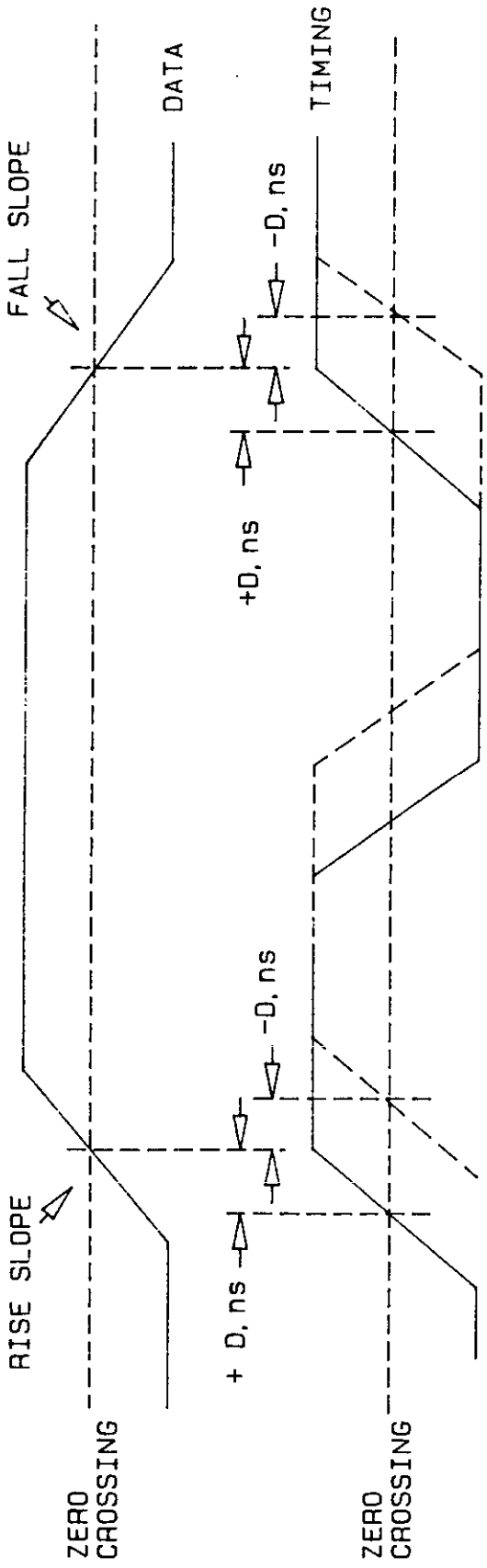


FIGURE 7 . Delay measurement (clock/data phasing).

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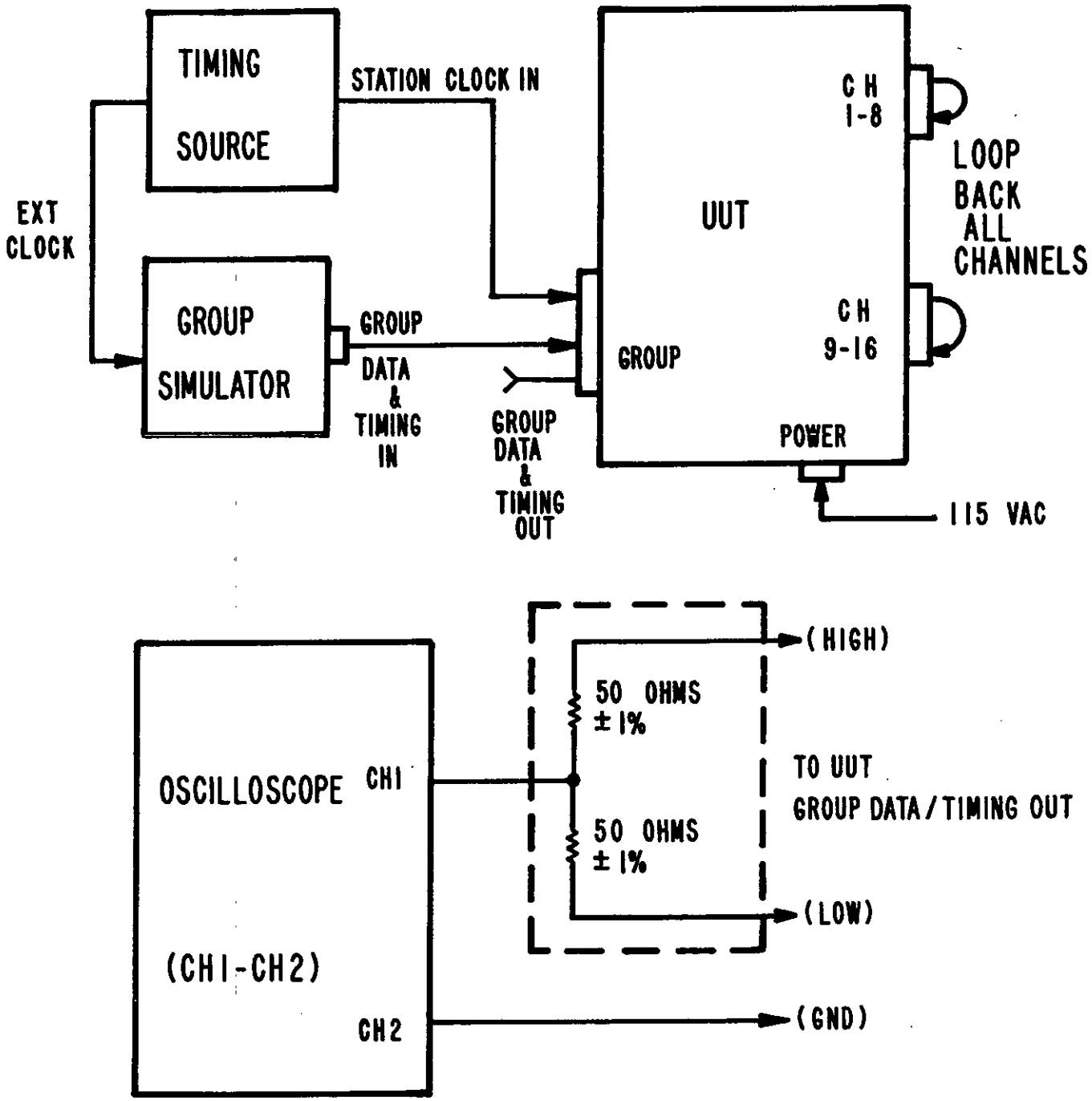


Figure 8. Offset.

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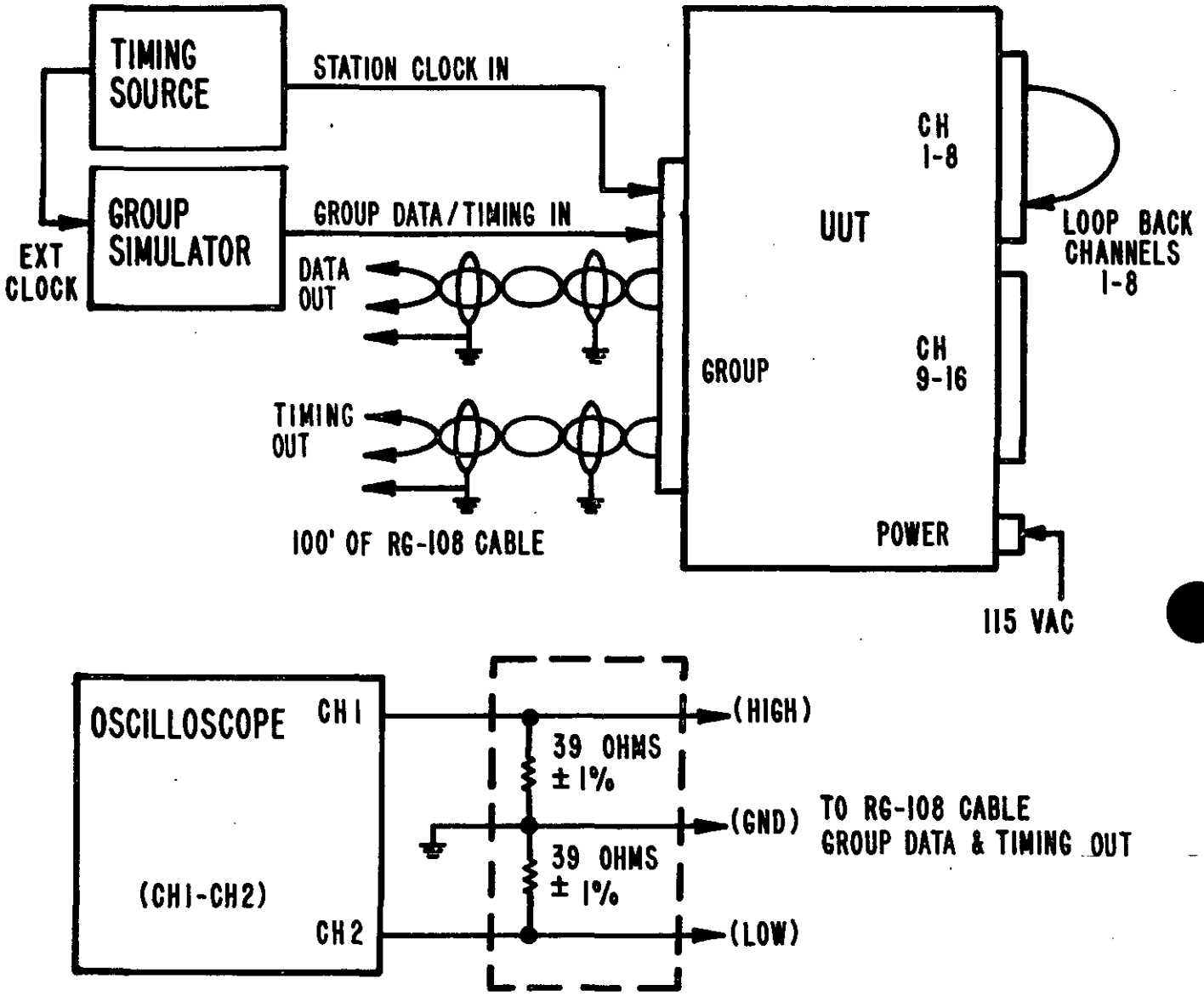


Figure 9. Slew rate.

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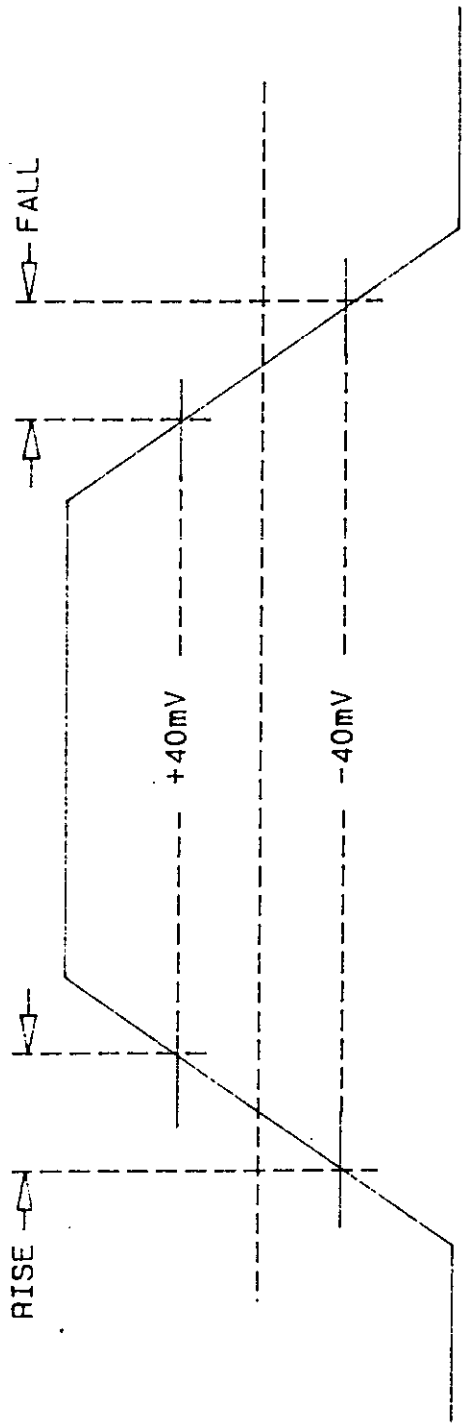


FIGURE 10. Rise and fall slopes (slew rate).

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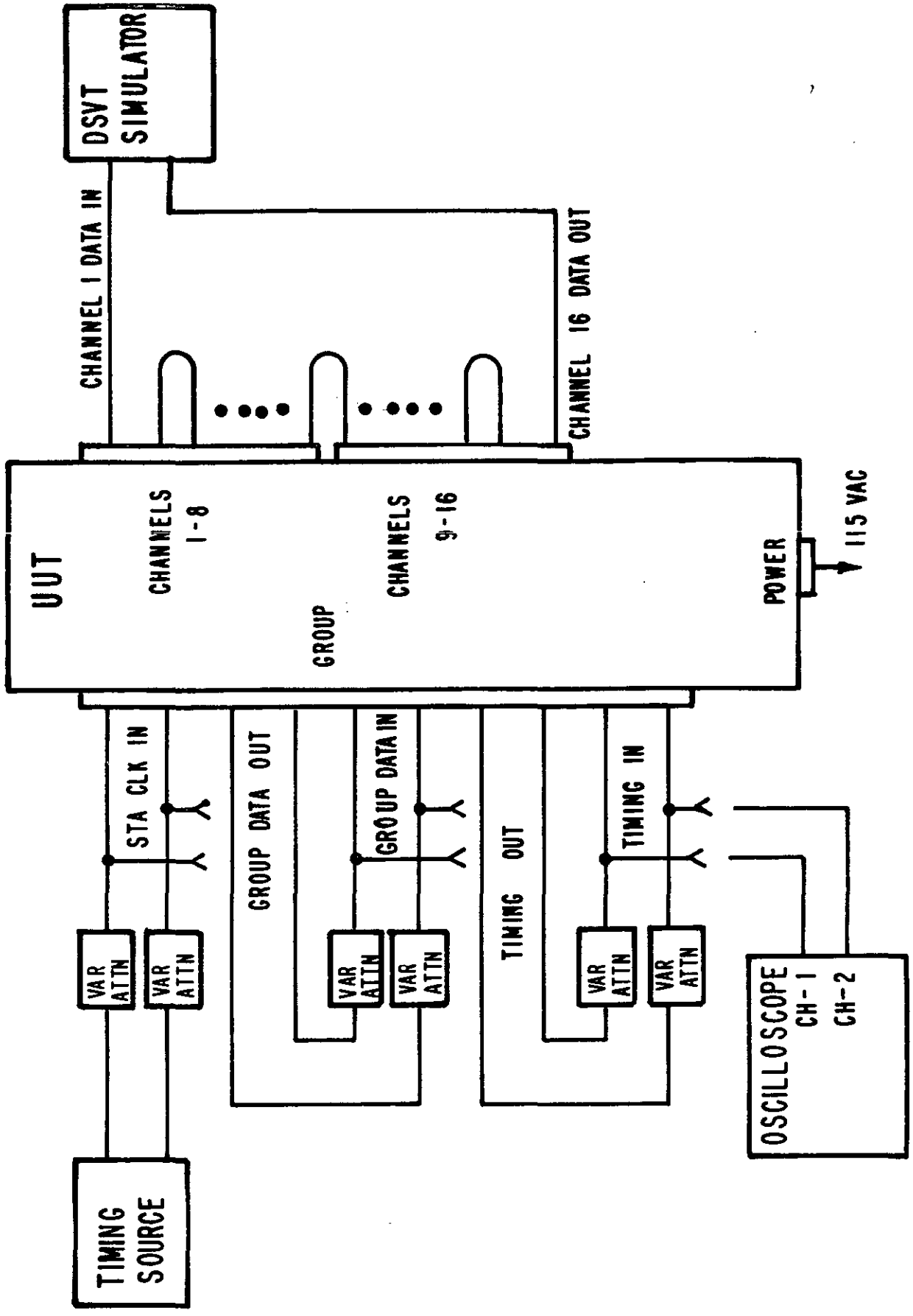


Figure 11. Receiver sensitivity.

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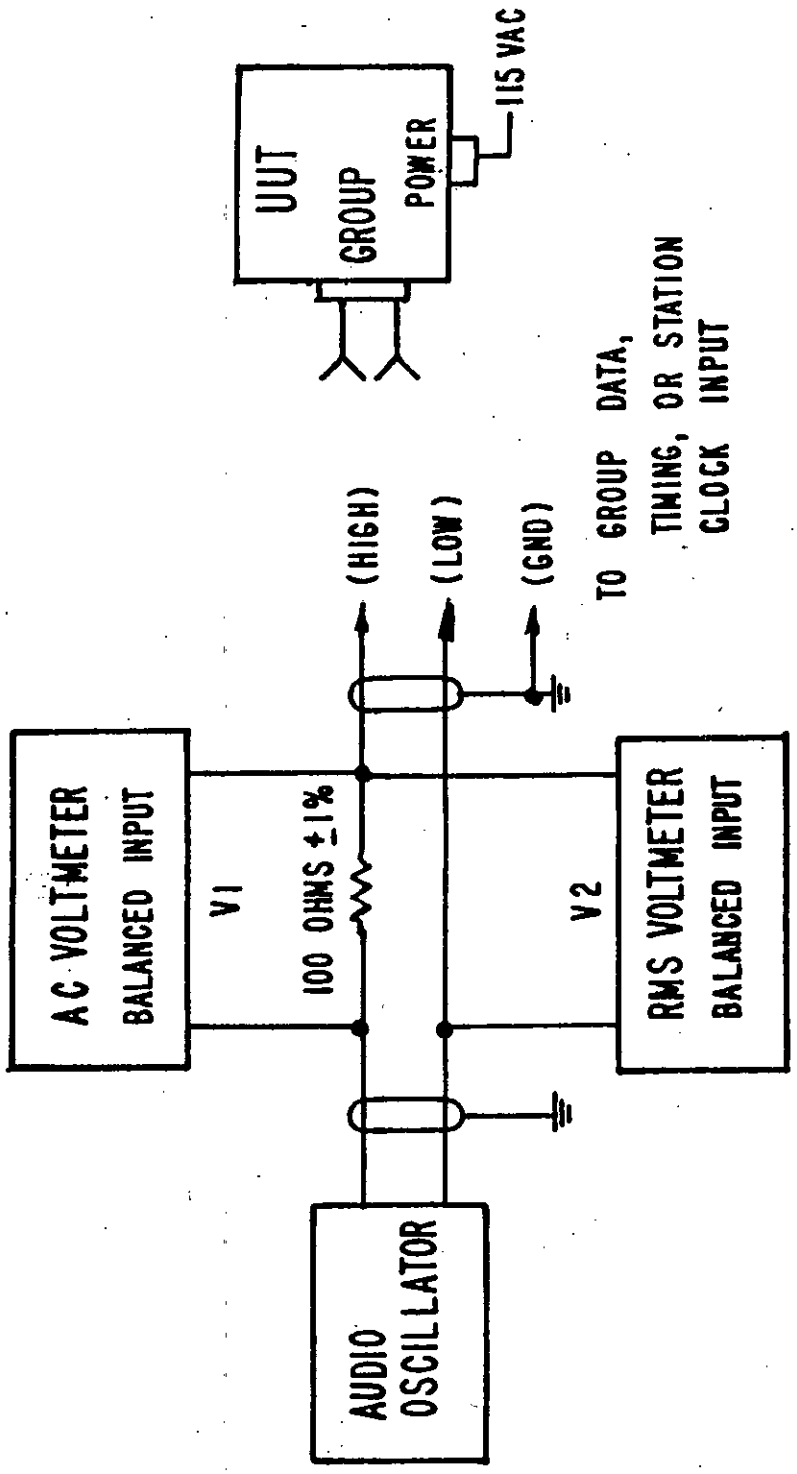


Figure 12. Input load impedance.

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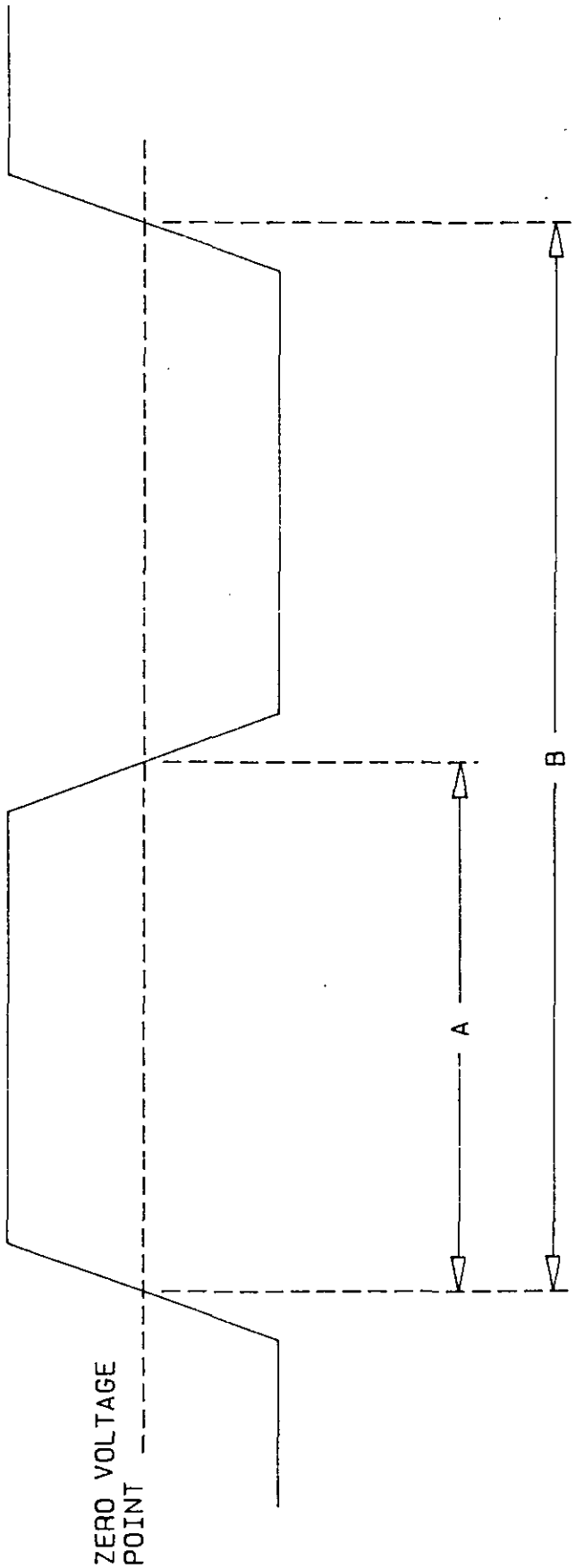


FIGURE 13 . Clock asymmetry.

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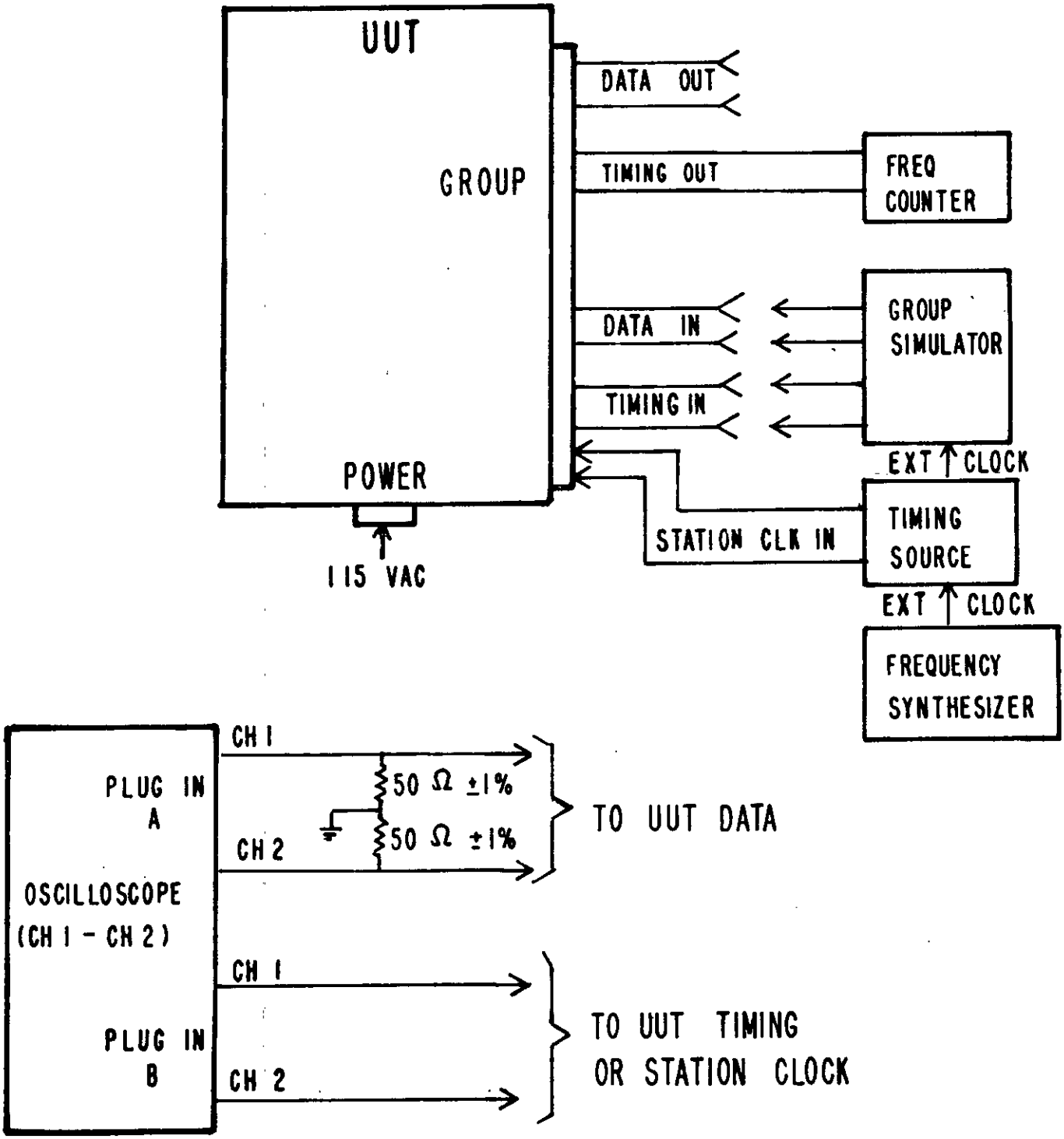
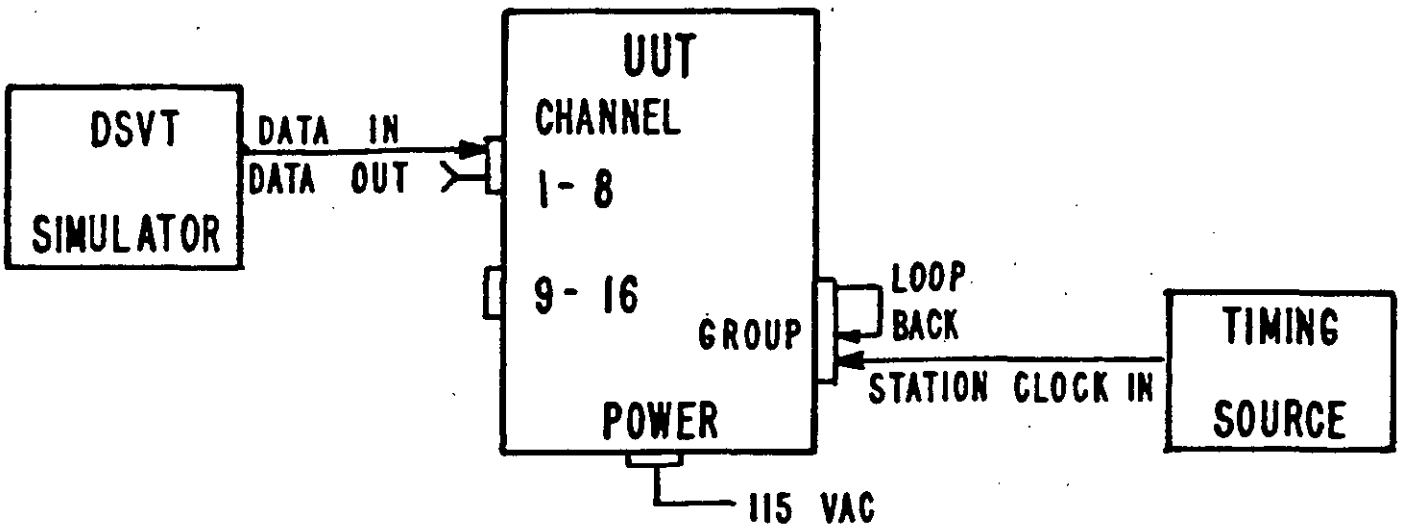


Figure 14. External clock data phasing.

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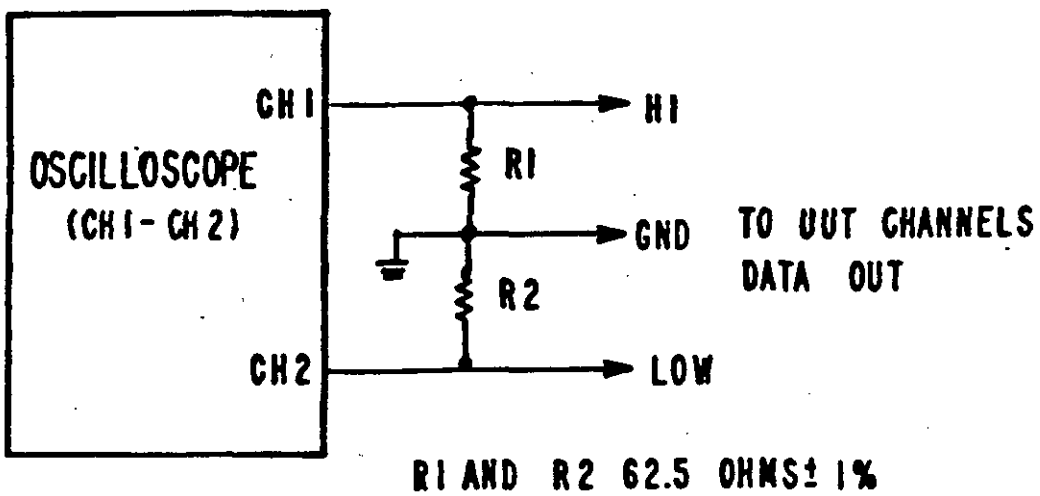


Figure 15. Output voltage and wave-shape.

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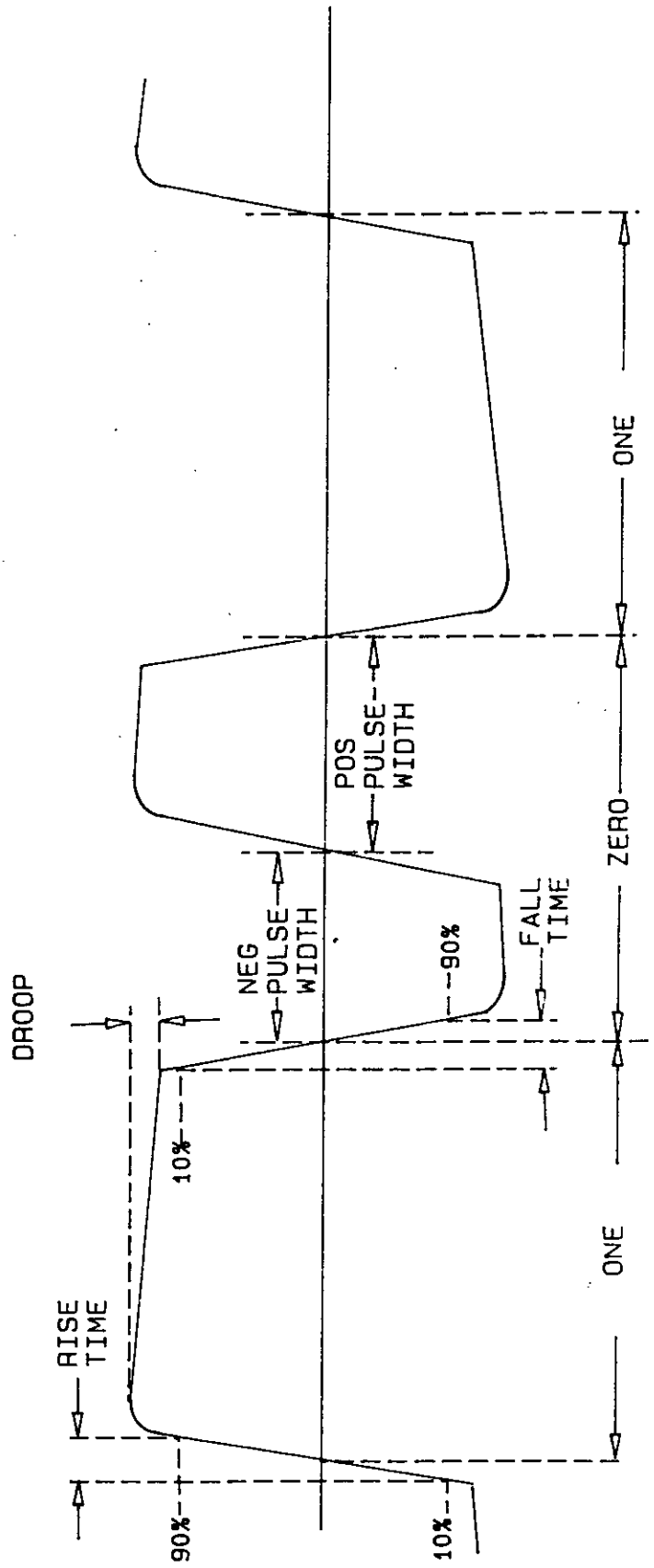


FIGURE 16. Diphase waveform (digital zero).

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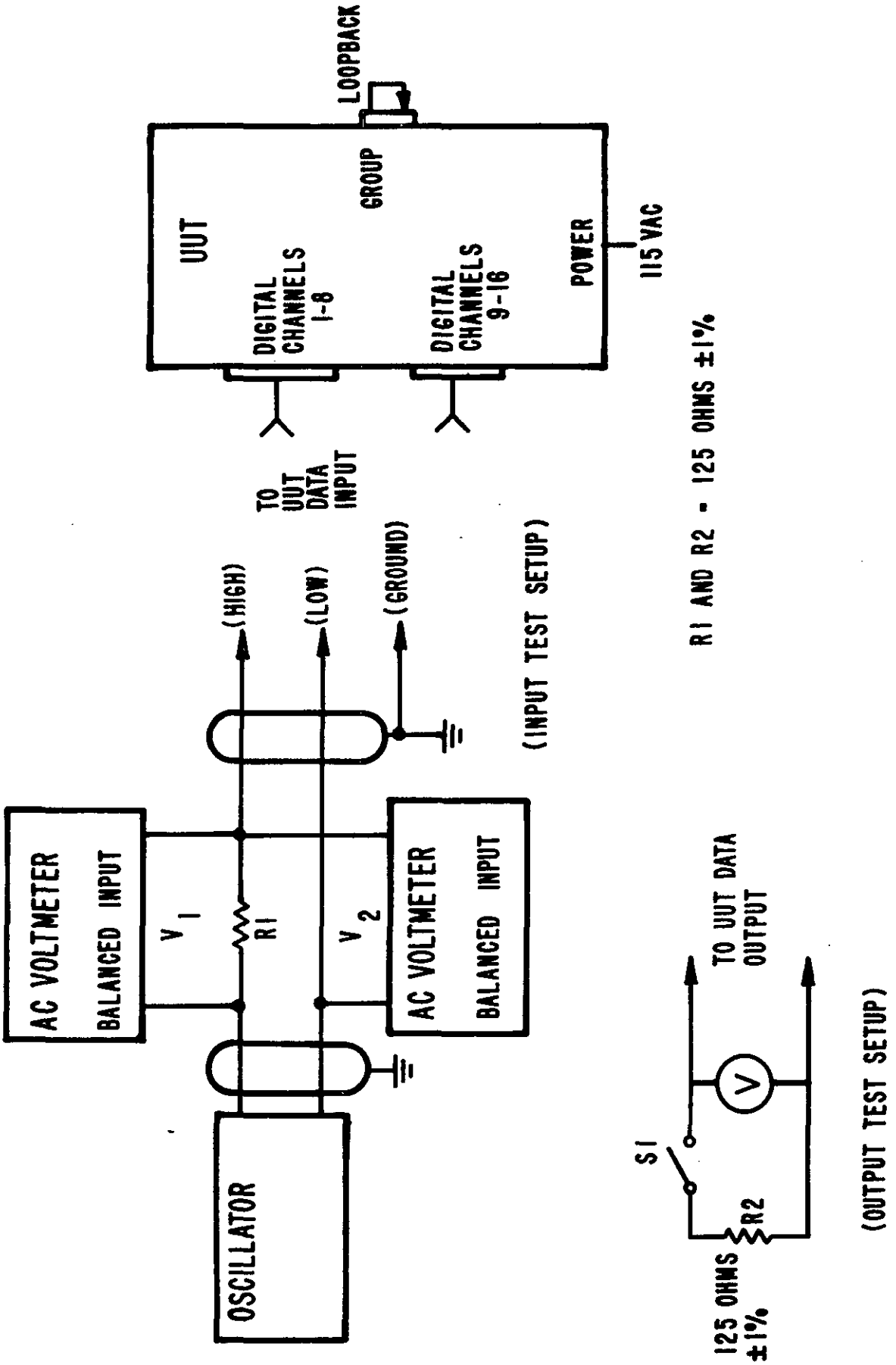


Figure 17. Input and output impedance.

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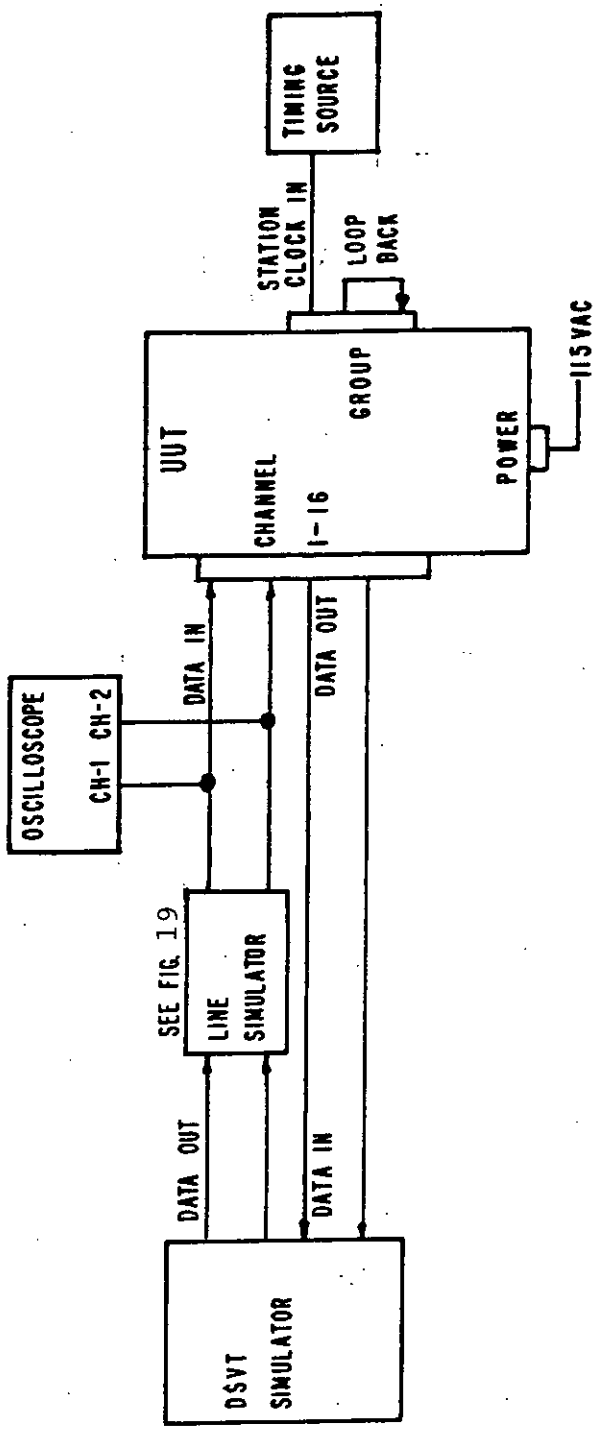


Figure 18. Input voltage.

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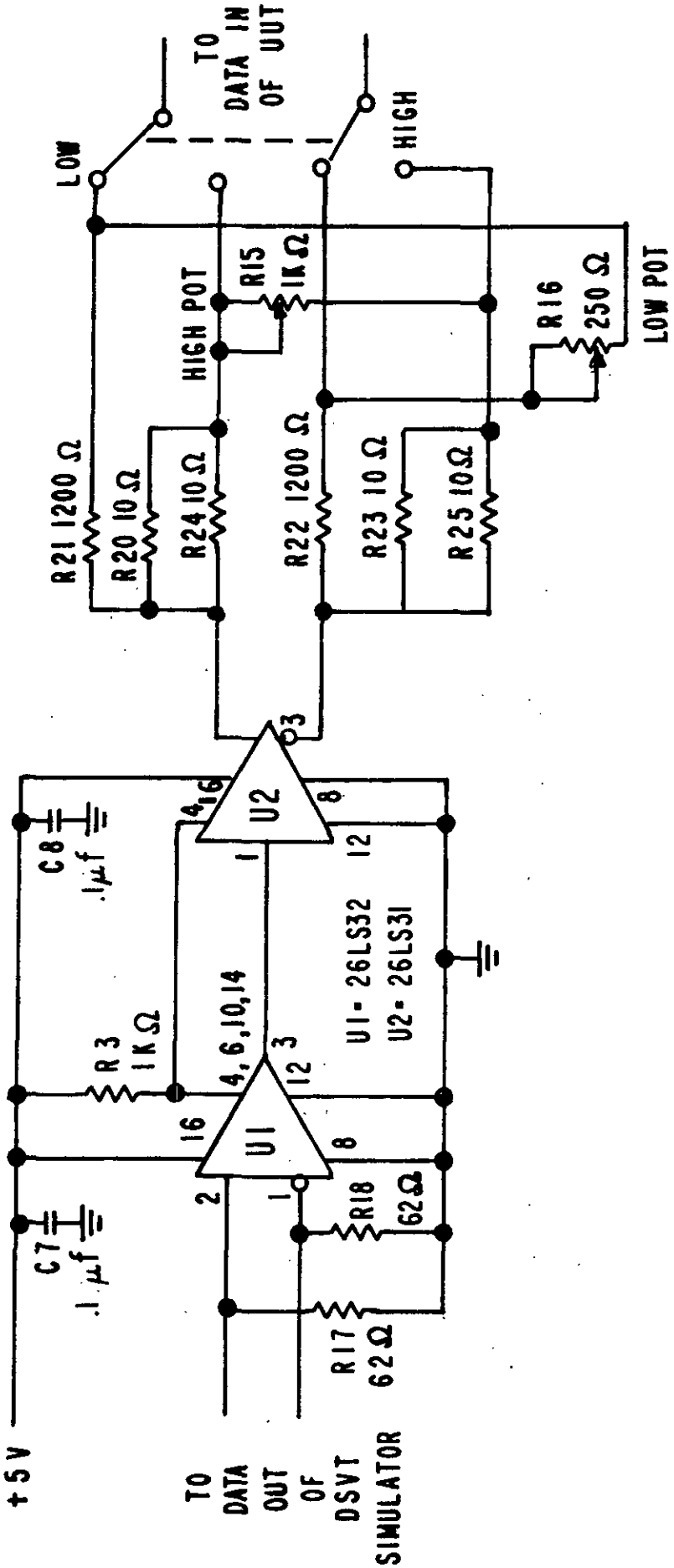


Figure 19. Line Simulator detail.

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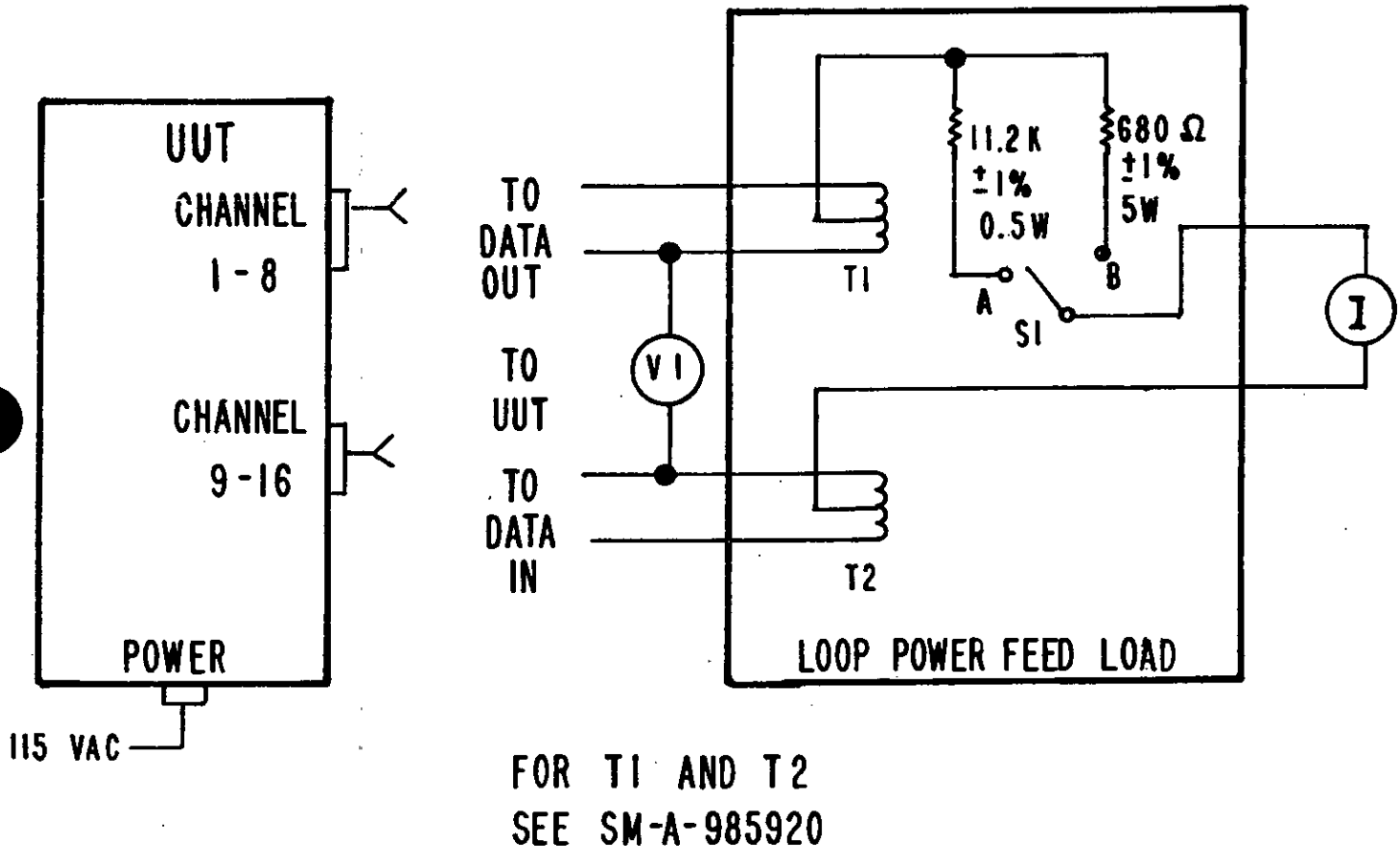


Figure 20. Loop power feed load.

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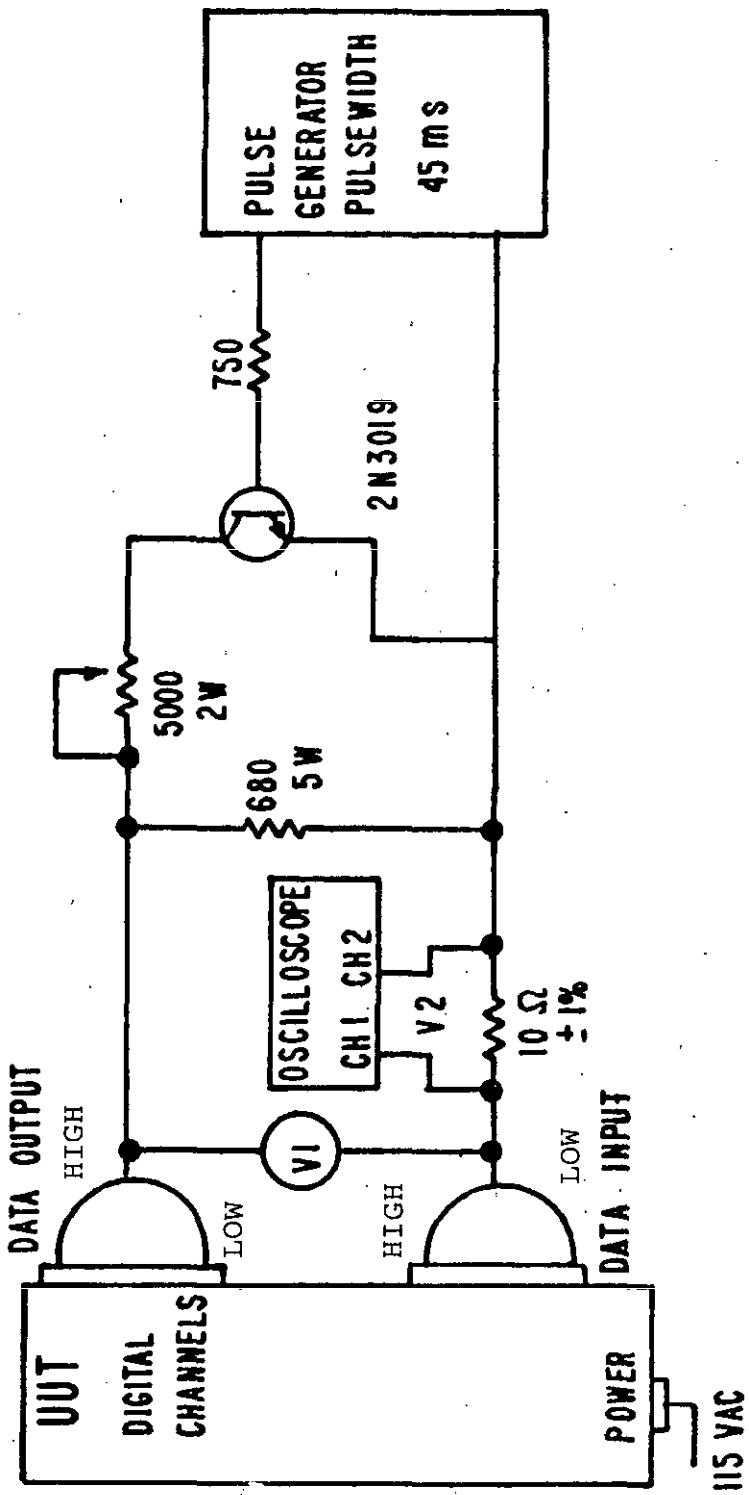


Figure 21. Loop power surge.

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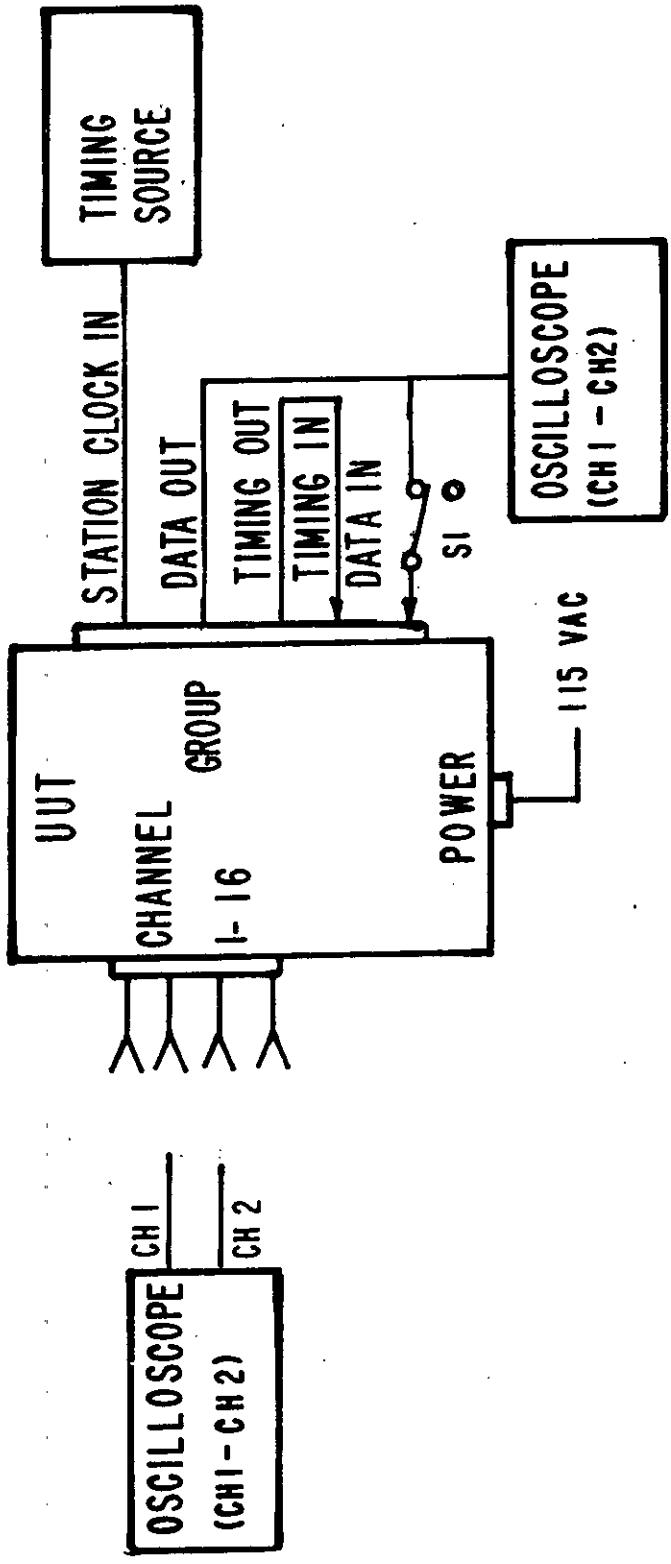


Figure 22. Idle loop.

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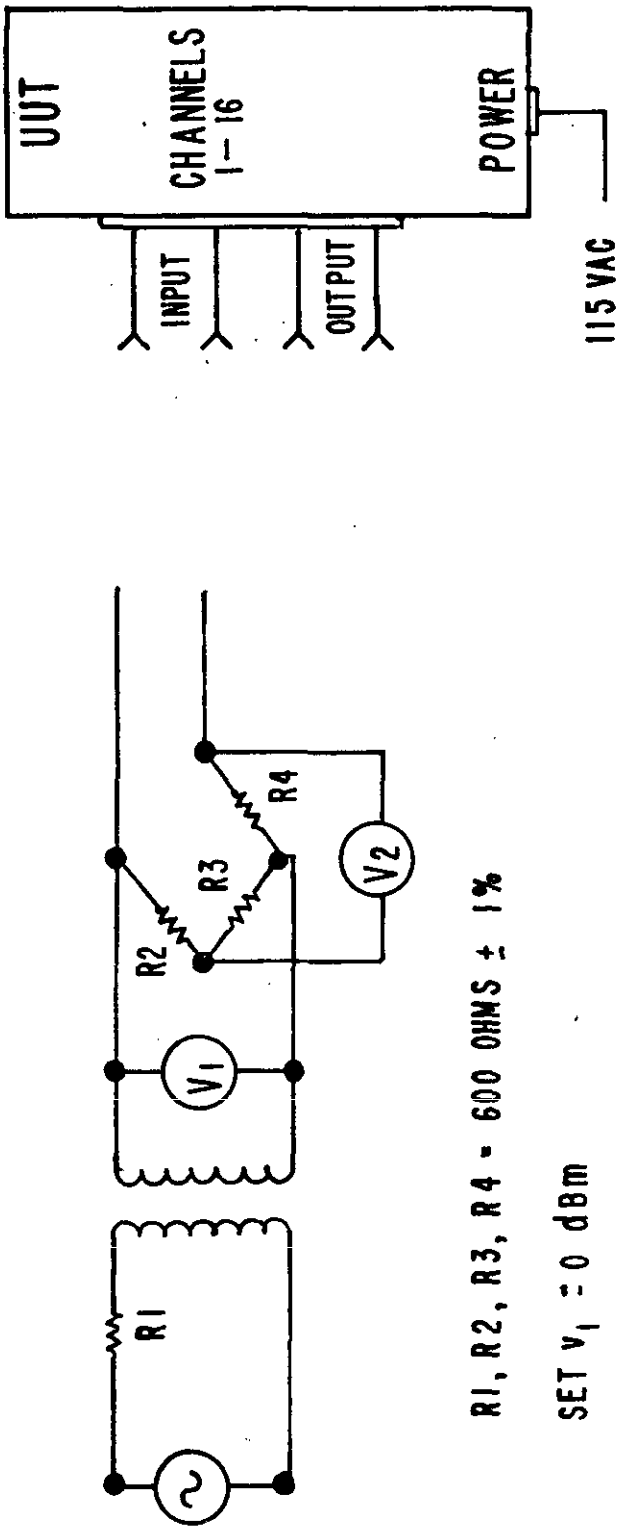


Figure 23. Impedance (return loss).

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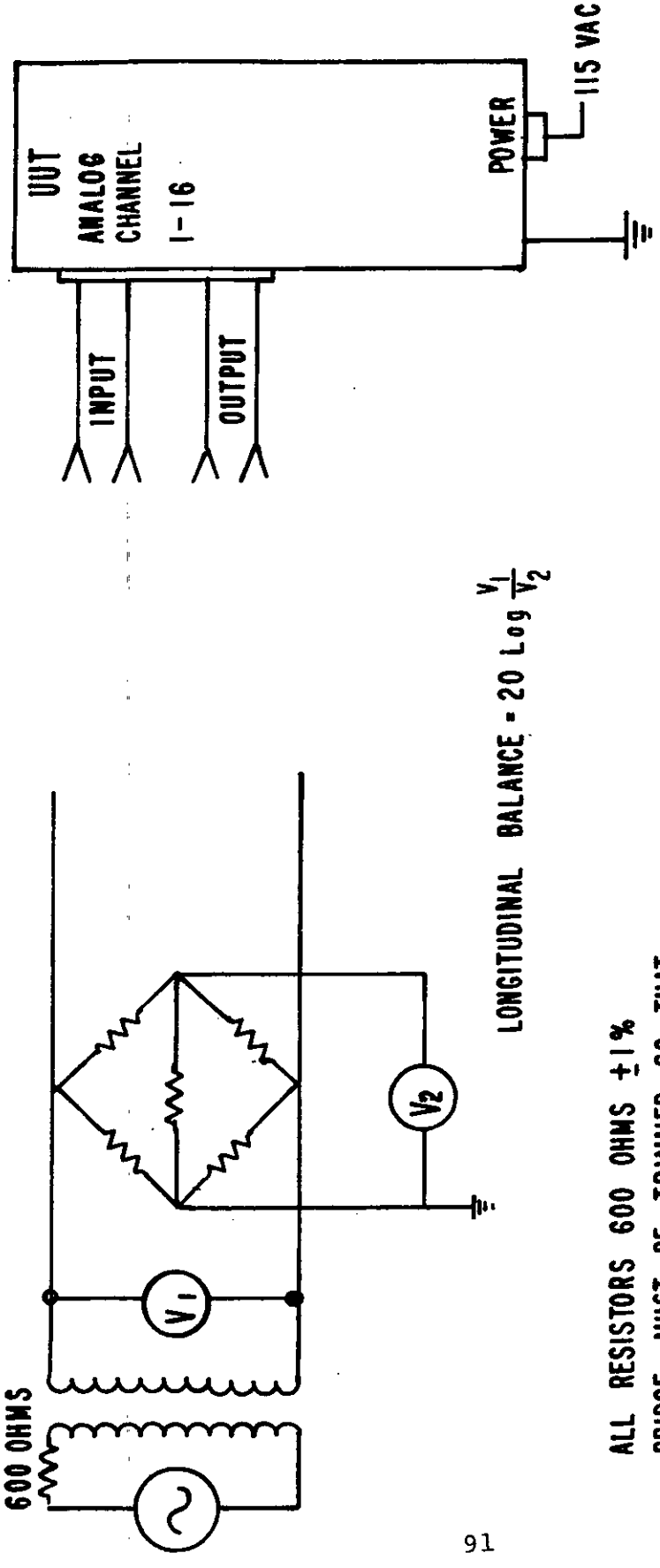


Figure 24. Longitudinal balance.

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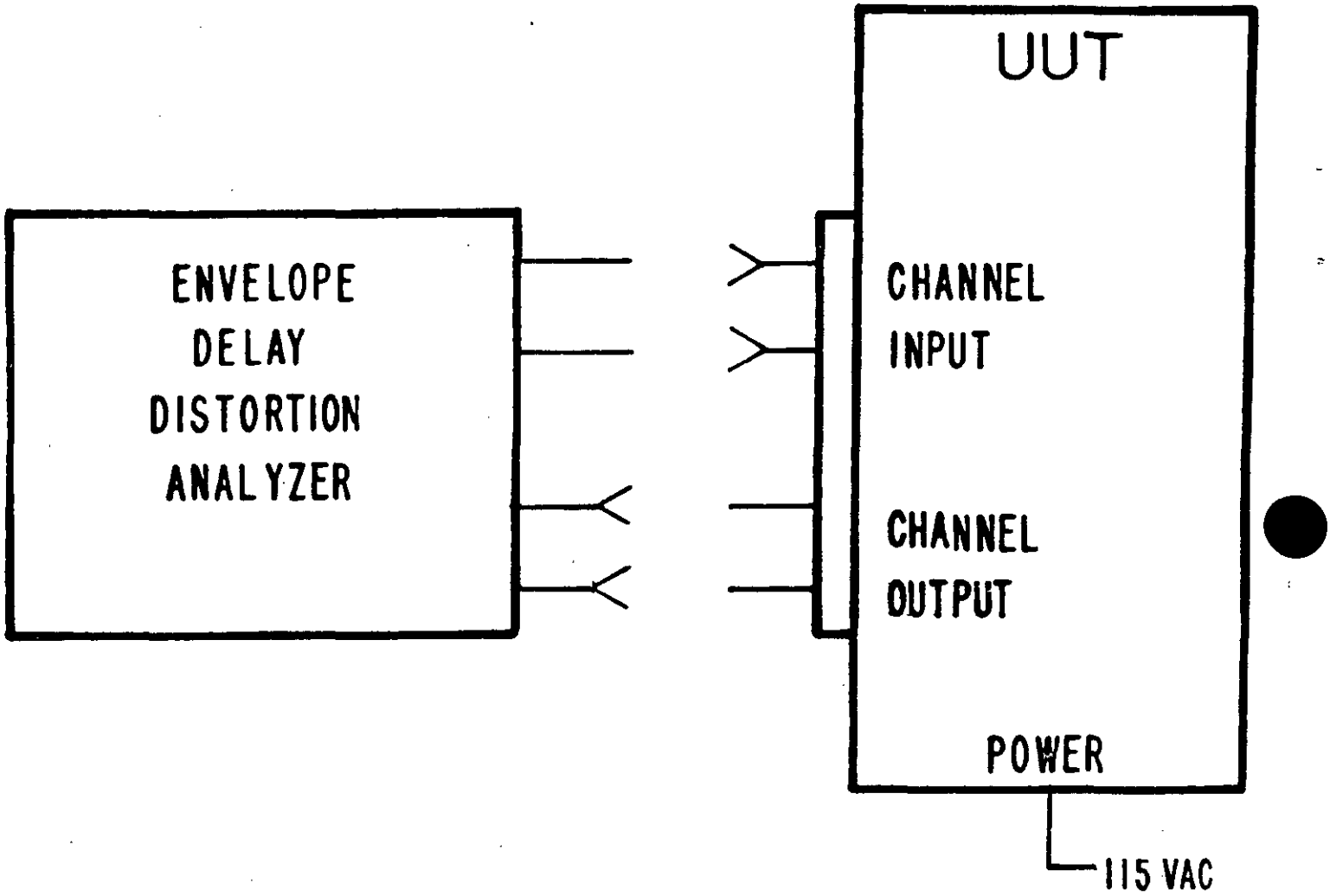


Figure 25. Envelope delay distortion.

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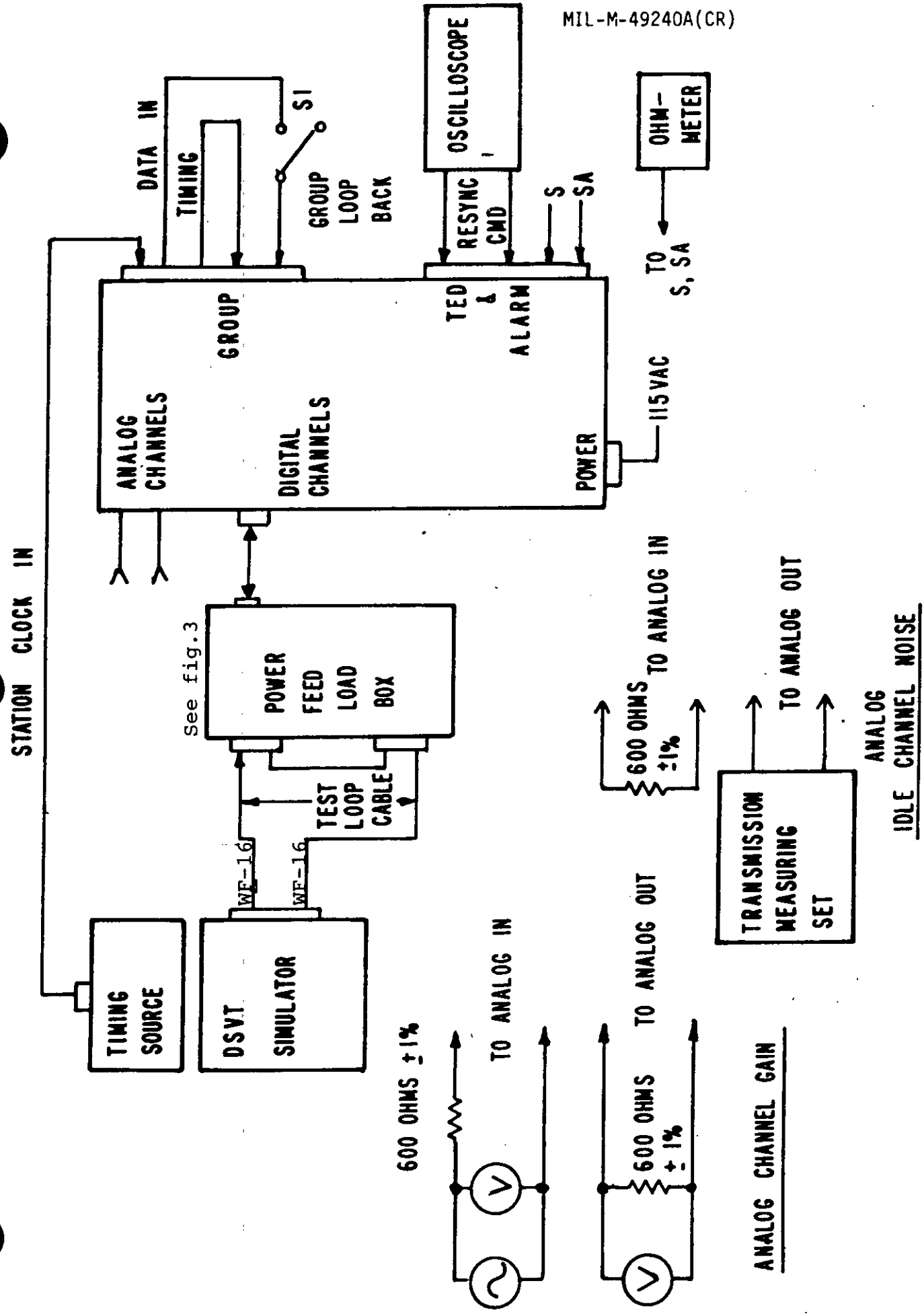


FIGURE 26. Idle channel noise; BITE and alarms; operational checks.

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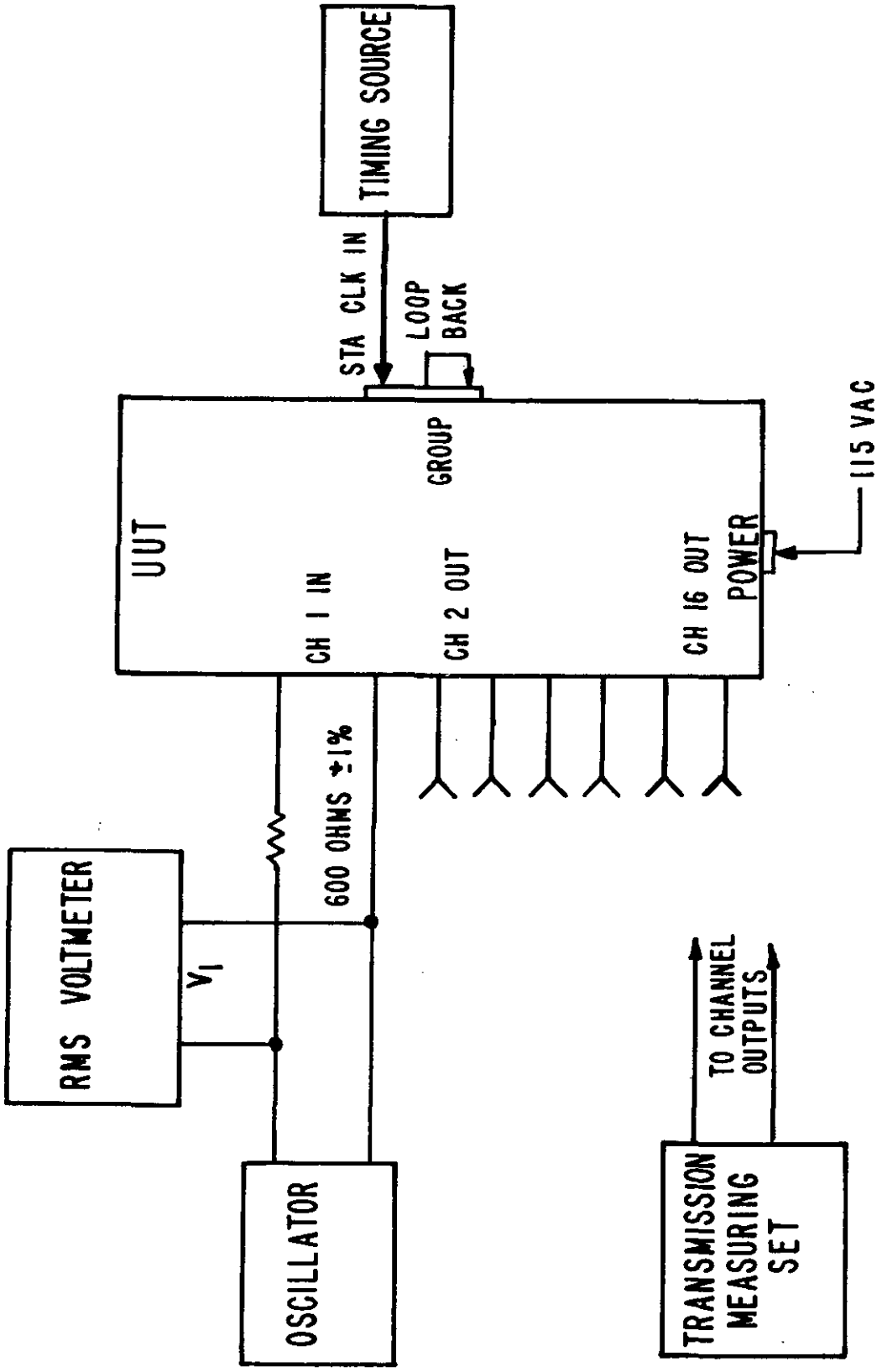


Figure 27. Crosstalk.

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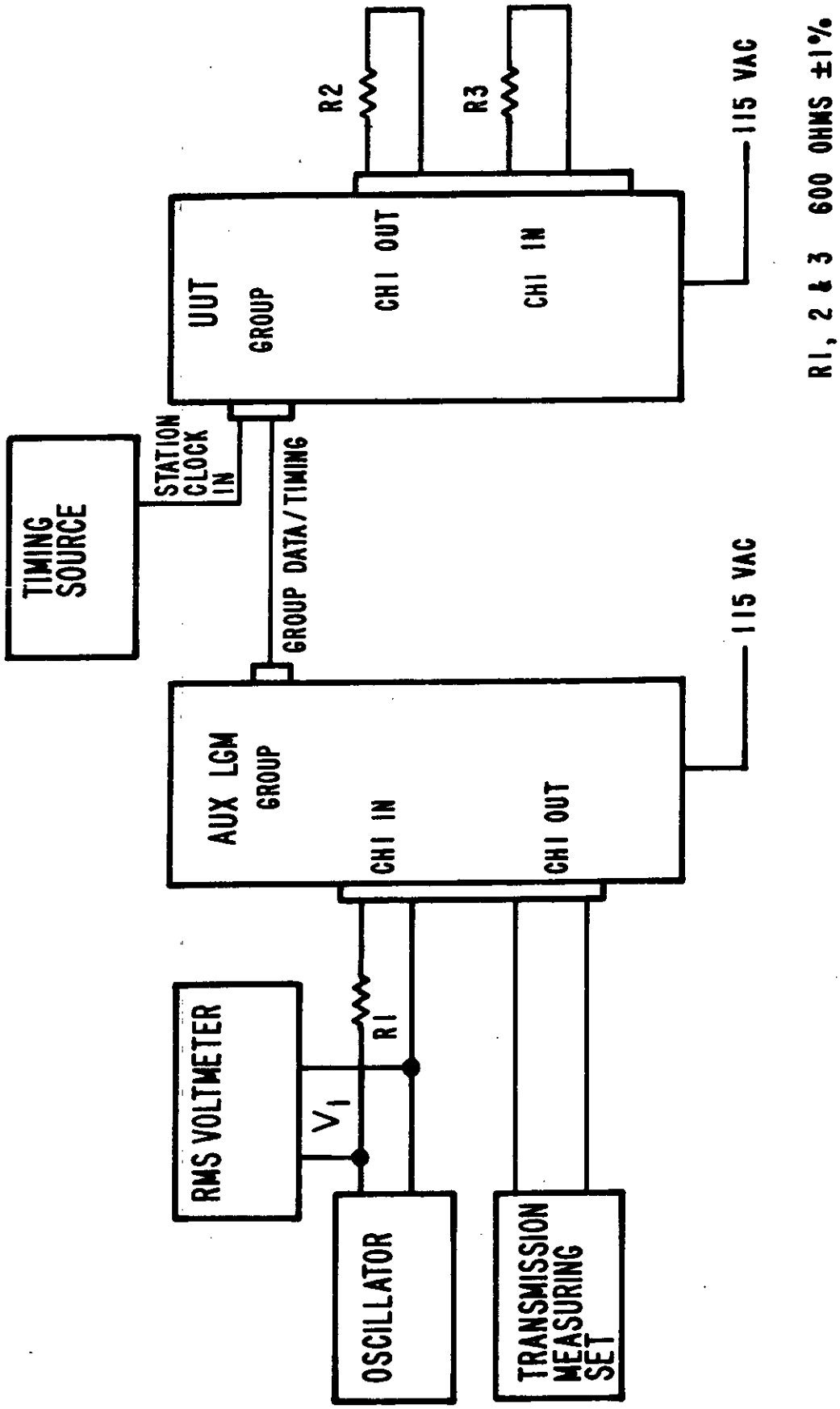


Figure 28. Crosstalk, second configuration.

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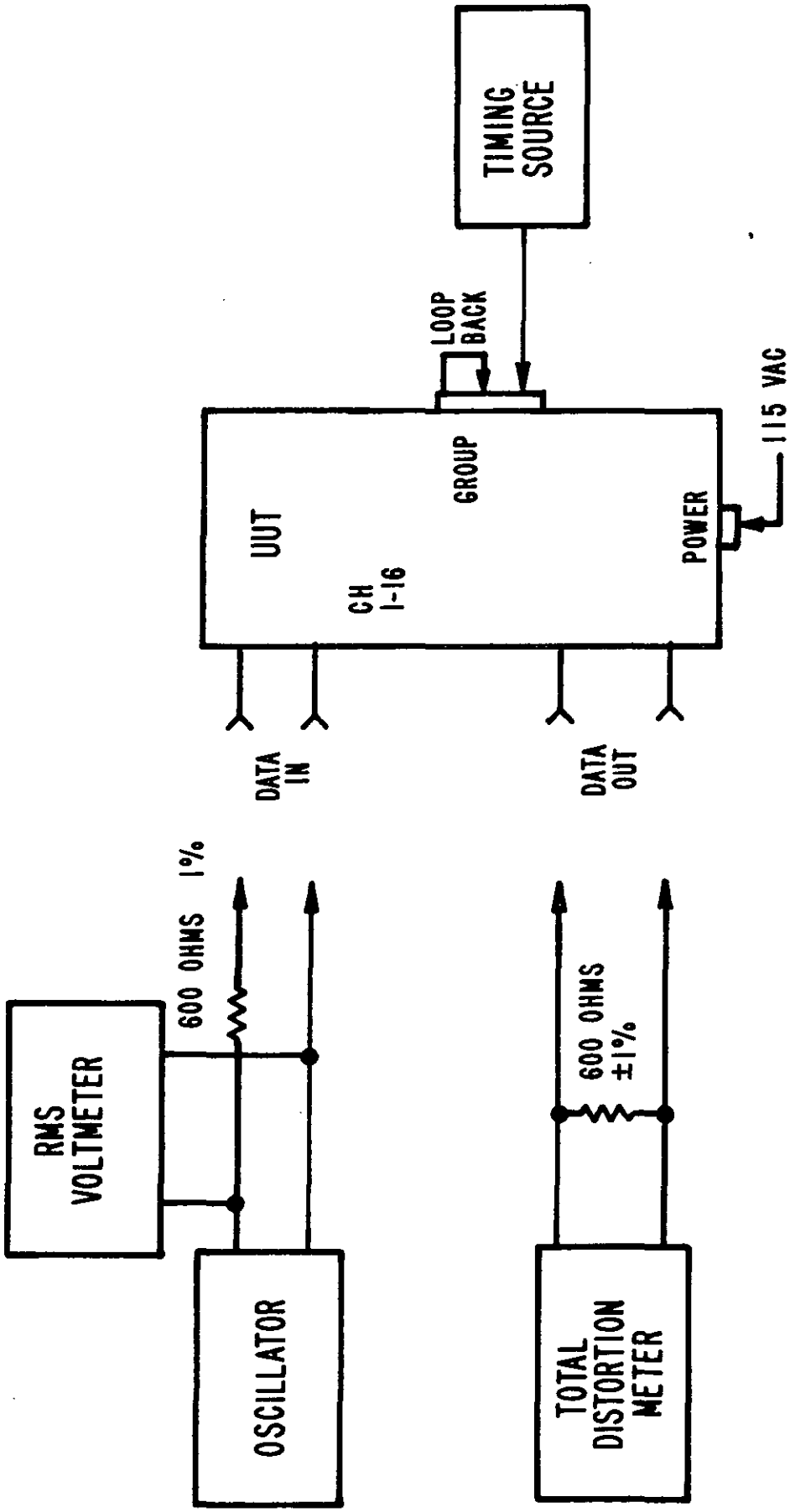


Figure 29. Peak clipping.

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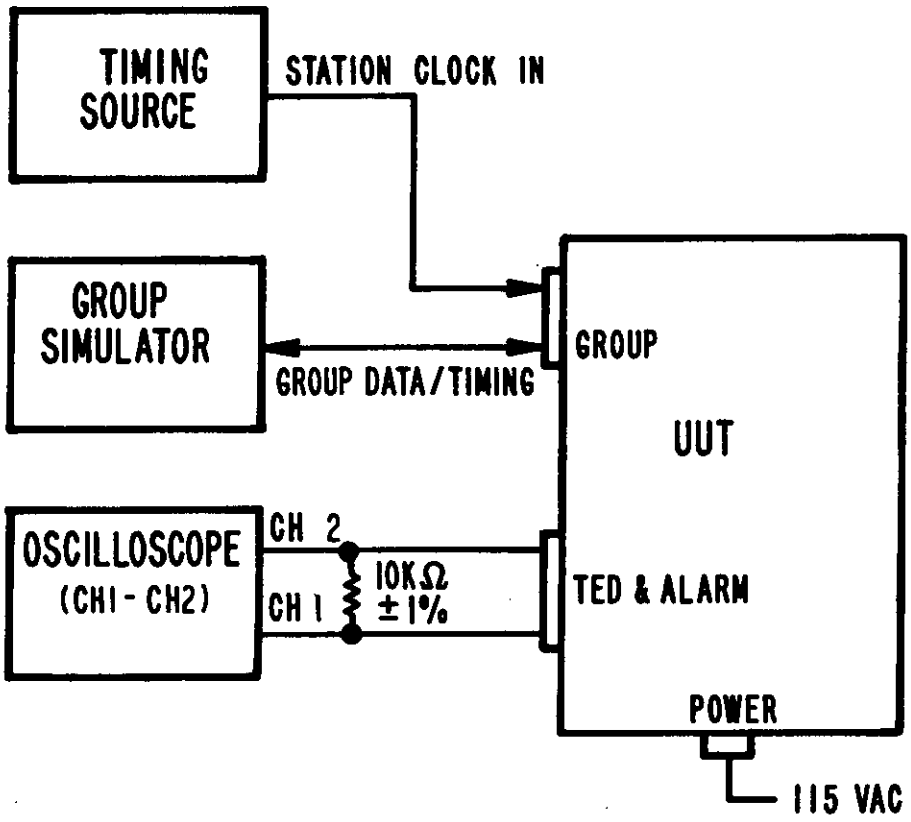
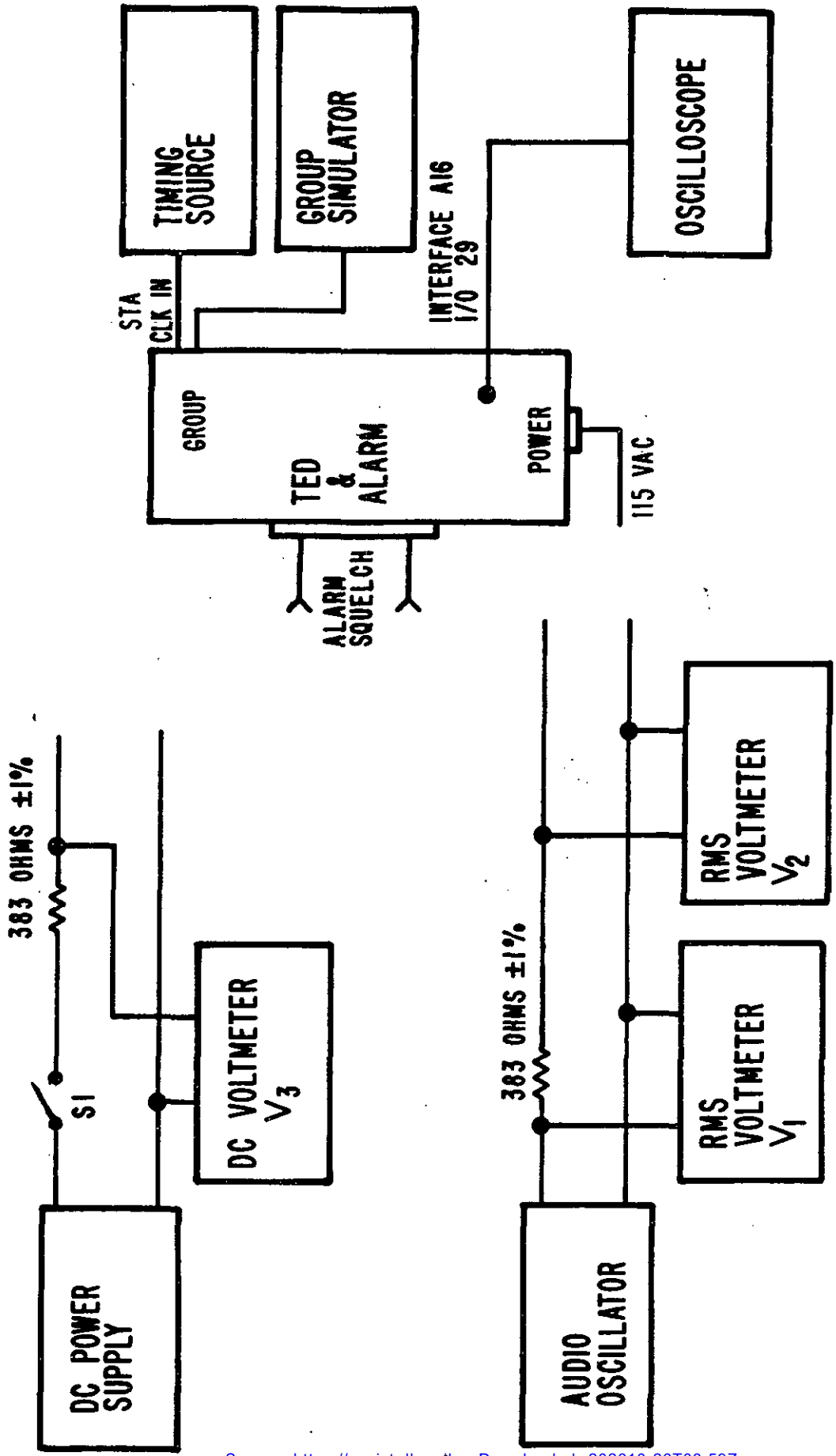


Figure 30. TED resync command.

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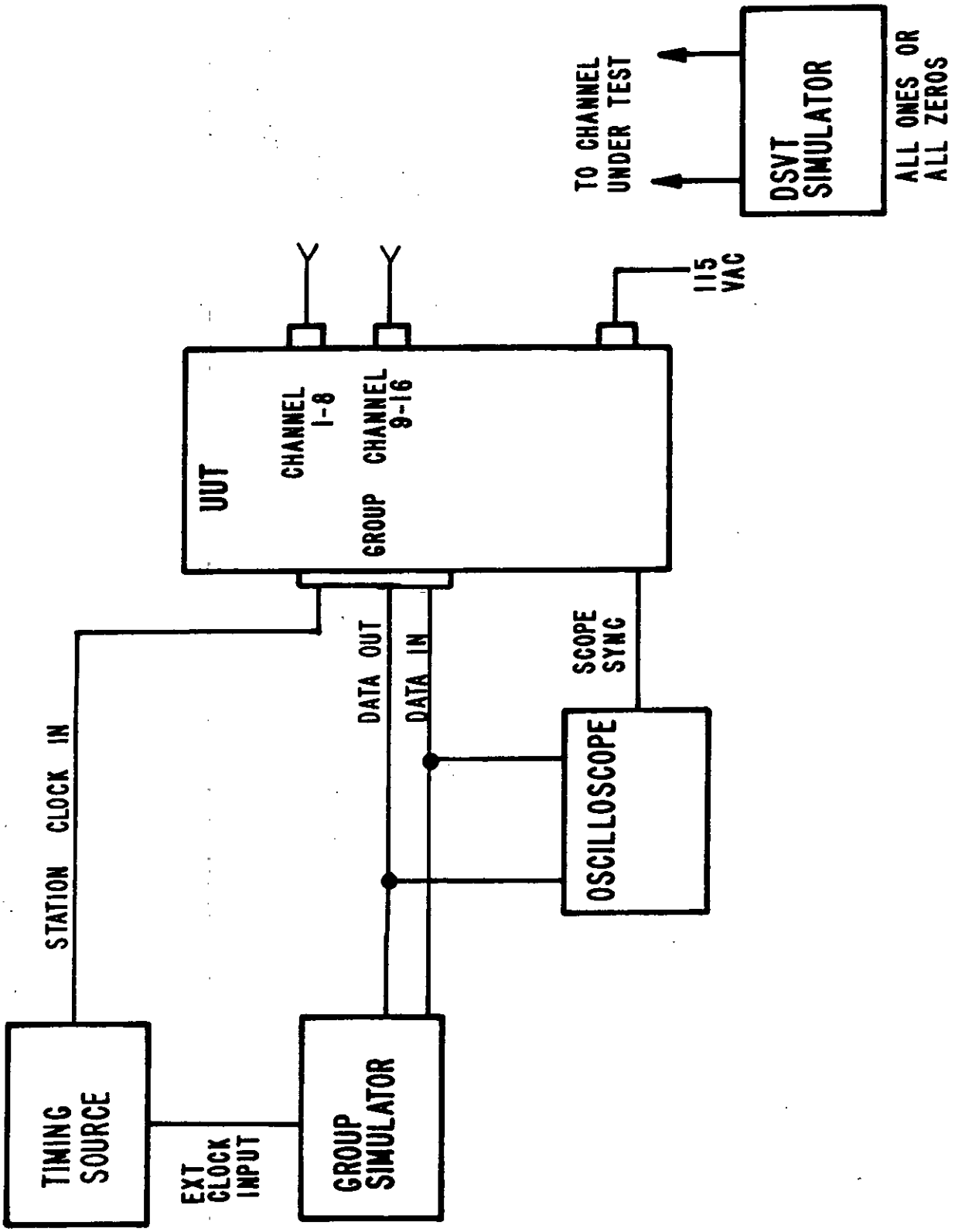


Figure 32. Format.

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GROUP RATE FOR CHANNEL RATE OF:		NO OF CHANNELS	PATTERN
16 kb/s	32 kb/s		
2	4	7	0 C1 C2 C3 C4 C5 C6 C7 0 C1 C2 C3.....
3	5	8	0 C1 C2 C3 C4 C5 C6 C7 C8 0 C1 C2 C3.....
4	6	15	0 C1 C2 C3 C4 C5 C6 C7 C8 C9 C10 C11 C12 C13 C14 C15 0 C1 C2 C3.....
5	7	16	0 SP C1 C2 C3 C4 C5 C6 C7 C8 C9 C10 C11 C12 C13 C14 C15 C16 0 SP C1 C2 C3..

TO/FROM SUB CH.

F	1	2	3	4	5	6	7	F	1

SP = SPARE CHANNEL SET TO 1

Figure 33. Output group patterns and overhead channels.

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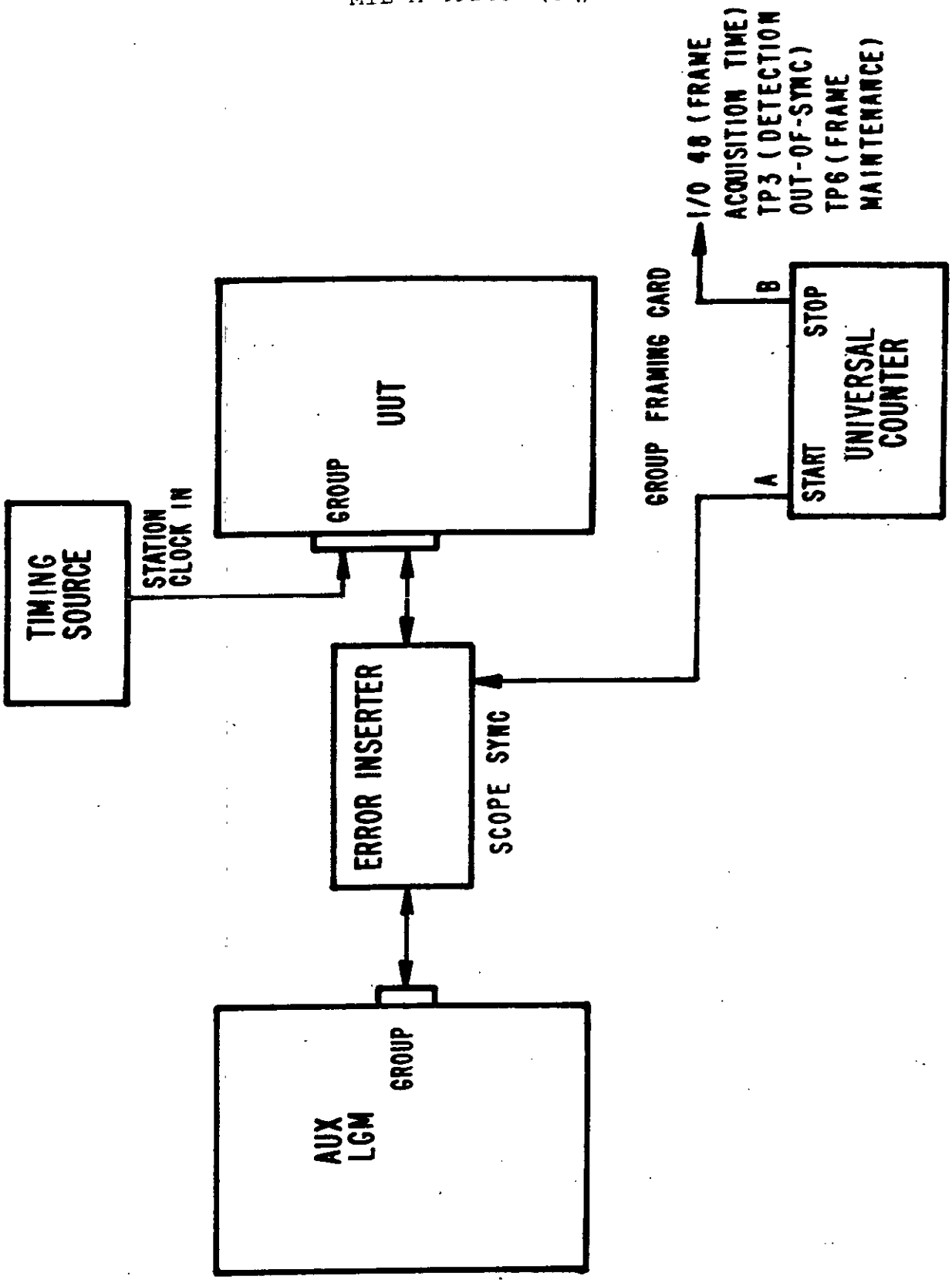


Figure 34. Frame acquisition time; maintenance; detection out-of-sync.

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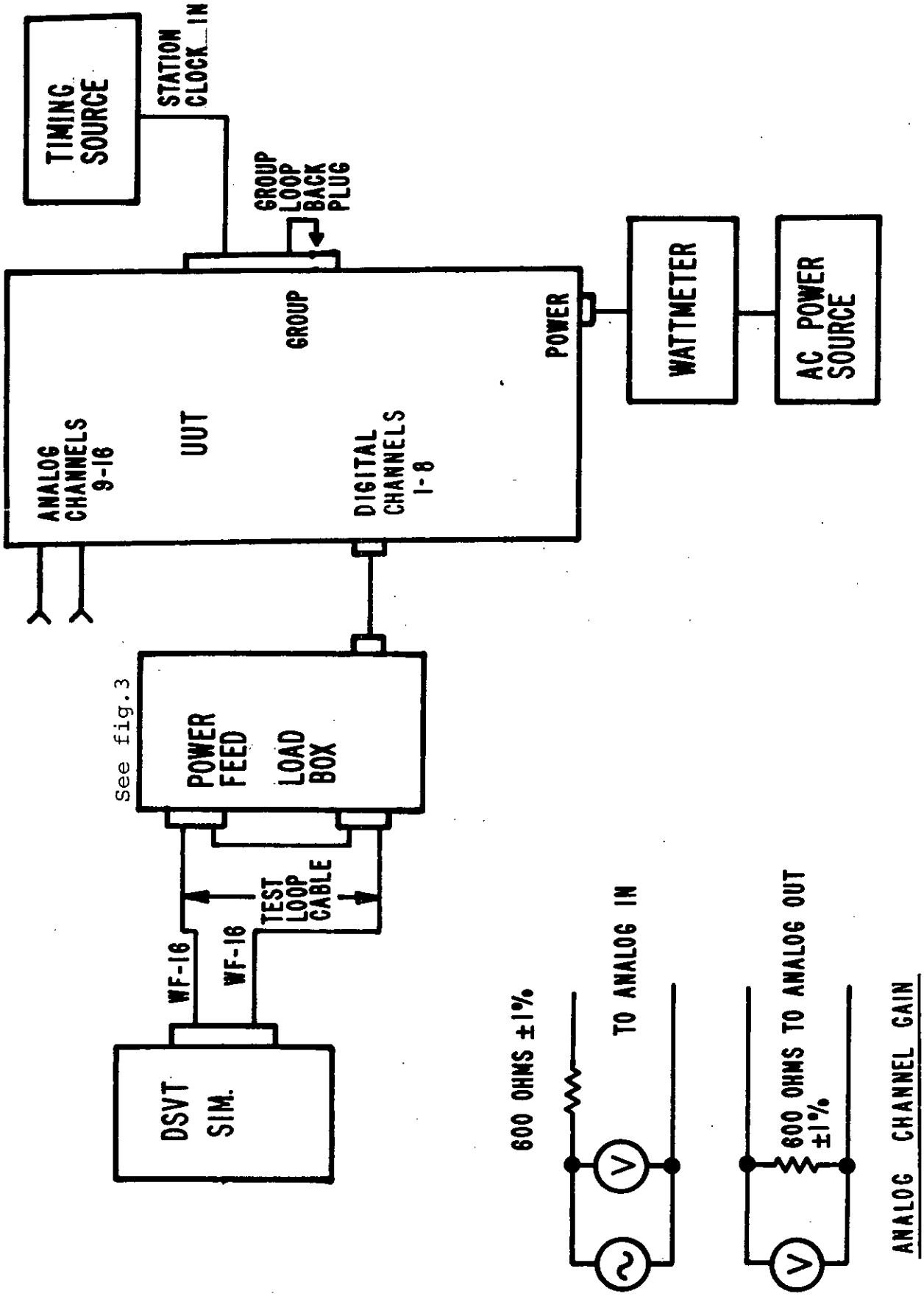


Figure 35. Power variation.

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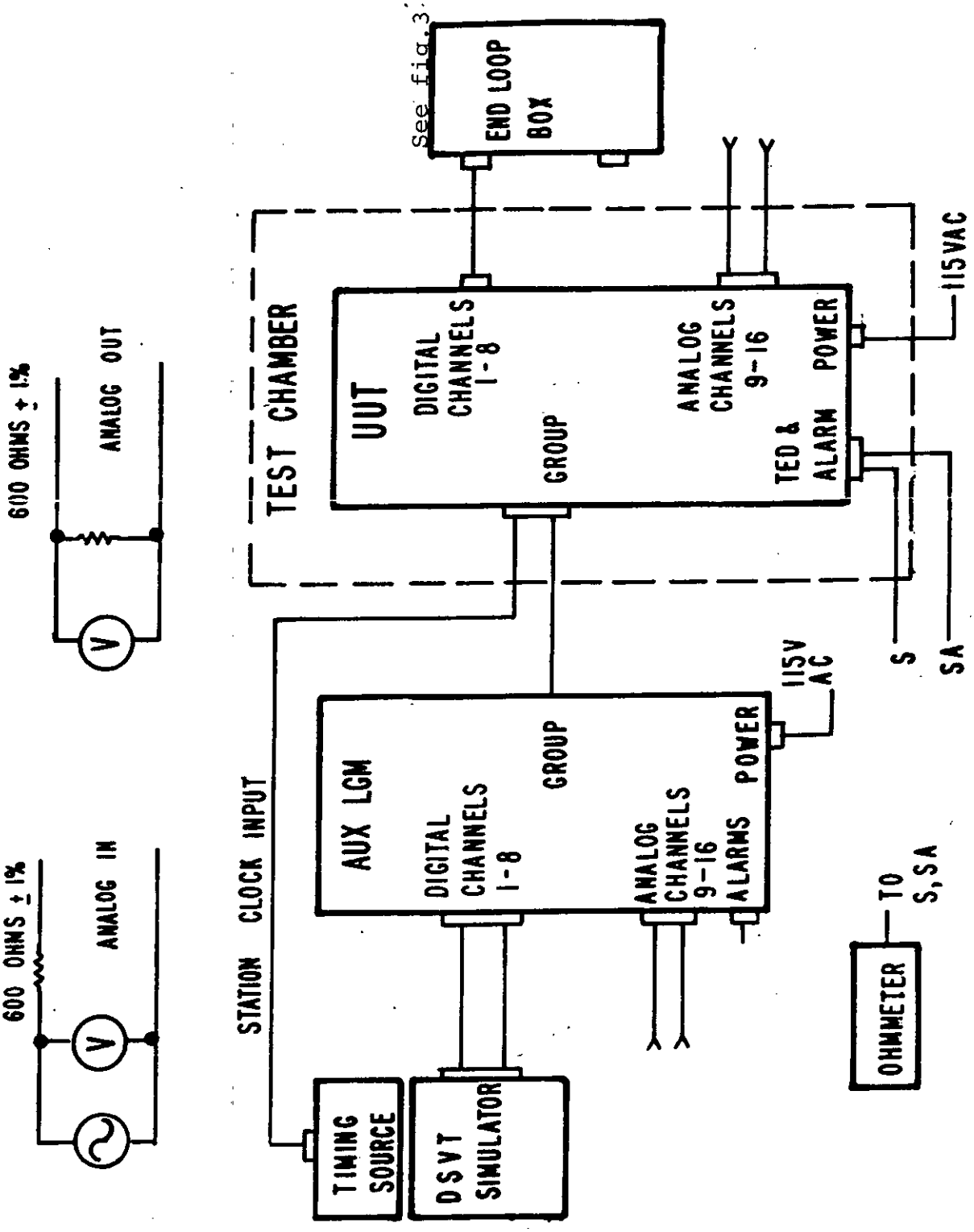


FIGURE 36. Environmental operational tests.

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